

TelcoX Customer Segmentation

Finding the customers who are ready to upgrade

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About this case study

TelcoX is a fictional telecom company. All data in this project is synthetic and no real customer information is used. The report combines the business analysis, the data work, the segmentation results, the marketing recommendations, and the measurement plan.

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Executive Summary

TelcoX wants to send each customer promotions that fit how they actually use their phone, and to find the customers most ready to move to a higher plan. This case study groups 15,000 customers by behavior, ranks the groups by upgrade potential, recommends an offer for each, and sets out a fair test to prove the approach works. TelcoX is fictional and all data is synthetic.

Key results

Result	Finding
Customer segments	Five, built from usage, spending, and promotion behavior
Customers who have outgrown their plan	7,206 (48 percent of all customers)
Best place to start	Heavy Data Users: 99 percent use more data than their plan allows
Stability	At least 99.3 percent of customers kept their segment in 20 subsample tests
Honest caveat	The segments overlap, so they are a practical way to divide customers, not natural separate groups

The five segments

Segment	Share	Upgrade rank	Main offer	Channel
Heavy Data Users	12.3%	1	Smallest plan that covers their data, with a first-month incentive	Mobile app
Long-Term Voice Users	8.1%	2	Free plan check-up for a plan that covers their calls	Shop, with SMS invite
Deal-Seeking Prepaid Users	18.3%	3	Bonus data on a larger monthly bundle	USSD
Steady Postpaid Households	34.2%	4	Home broadband and mobile bundle	Shop
Light Prepaid Users	27.0%	5	Plan with more call minutes, only if over the limit	USSD

How success will be measured

- **Four main measures:** upgrade rate, response rate, ARPU change, and cost per extra upgrade, plus guardrails for opt-outs, complaints, and disconnections.
- **A fair test:** eligible customers are split at random into a targeted half and a control half that gets the usual offer. Segments 2 to 5 are pooled (5,786 customers) so the test can detect a 45 percent lift in upgrades over three months.
- **Proposed targets:** upgrades at least 45 percent above control, response at least 25 percent above control, and cost per extra upgrade at or below break-even.

Next steps

1. Confirm the targets and the open questions in the implementation plan.
2. Run the pilot, starting with Heavy Data Users.
3. Review at three months, and extend to six if the result is unclear.
4. Refresh the segments every quarter.

Business Requirements Document

Project name	TelcoX Customer Segmentation
Document	01 Business Requirements Document (BRD)
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)

Note on this case study. TelcoX is a fictional company. All data used in this project is public or synthetic. No real customer information is used.

1. Purpose of This Document

This document explains what TelcoX wants to achieve with customer segmentation, why it matters, and what the project must deliver. It is the starting point for all other project documents. It is written so that business and technical readers can both understand it.

2. Background

TelcoX is a mid-sized telecom company. It sells mobile voice, SMS, mobile data, and home broadband to several million customers. Customers are on either prepaid plans (pay before use) or postpaid plans (pay after use).

TelcoX operates in one country and reaches customers through retail shops, a mobile app, and a self-service channel such as USSD.

The telecom market is very competitive. Customers can switch to another provider easily, and competitors often run strong promotions.

Today, TelcoX marketing mostly sends the same offers to almost everyone. This approach has three problems:

- Response to campaigns is low.
- Promotion spending is high.
- Many customers get offers that do not match how they use their phone or internet.

3. Business Problem

TelcoX wants to target different groups of customers with promotions and services that suit them. It especially wants to find the groups most likely to upgrade or switch to a higher-tier plan.

At the moment:

- Average revenue per user (ARPU) has not grown. ARPU is the average amount each customer pays per month.
- Many customers stay on basic plans even though their usage suggests a bigger plan would suit them better.
- TelcoX has no data-based customer segmentation. Groups are defined only by plan type or how long someone has been a customer.
- Marketing, sales, and product teams each use a different meaning for "high-value customer."

Key questions to answer

1. What different groups of customers exist, based on how they actually use TelcoX services?
2. Which groups are most likely to upgrade to a higher-tier plan?
3. What offers and services suit each group?

4. Project Goals

Business goals

#	Goal
B1	Increase the number of customers who upgrade to higher-tier plans.
B2	Improve how many customers respond to promotions.
B3	Reduce money spent on offers that do not suit the customer.
B4	Grow ARPU, especially in groups with unmet needs.
B5	Give all teams one shared view of customer groups.

Analysis goals

#	Goal
A1	Find a small number of clear customer segments.
A2	Describe each segment using demographics, behavior, and upgrade potential.
A3	Turn each segment into a recommended marketing action.

5. Scope

In scope

- Collecting and cleaning customer data (demographics, usage, and transactions).
- Grouping customers into segments using K-means clustering.
- Describing each segment in plain business terms.
- Estimating which segments are most ready to upgrade.
- Recommending an offer, channel, and timing for each segment.
- Defining how success will be measured.

Out of scope

- Building or changing TelcoX's live campaign or billing systems.
- Running real marketing campaigns.
- Setting new prices or designing new plans.
- Predicting customer churn (customers leaving) as a separate model.

Churn prediction is left out to keep the project focused on upgrade potential. It could be a future extension.

6. Stakeholders

Stakeholder	What they need
Marketing	Clear groups they can target, and campaign guidance.
Product and pricing	To know if current plans match real customer usage.
Sales and retail	Simple segment descriptions that shop staff can use.
Customer care	To understand common problems for each segment.
IT and data teams	A clear list of data needed and how results will be used.
Leadership	Proof that the work improves upgrades, ARPU, and campaign returns.

A fuller version of this table is in the next document, 02 Stakeholder Analysis.

7. Available Data

TelcoX holds three main types of customer data.

Data type	Examples
Demographic	Age group, gender, location (city or rural, region), account type (prepaid or postpaid), time as a customer, how they signed up.
Usage	Monthly call minutes, SMS count, mobile data used, share of off-peak use, roaming use, home broadband use.
Transaction	How often and how much they top up, bill payments, bundle purchases, past plan changes, past promotion responses, late payments.

The data sits in separate systems: billing, the customer database (CRM), network records, and the campaign tool. They share a customer ID but have not been combined, and data quality differs between them.

Some demographic details are missing for prepaid customers.

8. Business Requirements

These describe what the business needs from this project. Detailed user stories will follow in document 03.

ID	Requirement	Priority
BR1	The project must group customers into segments based on usage and behavior.	High
BR2	Each segment must have a clear name and description that non-technical staff can understand.	High
BR3	The project must show which segments are most likely to upgrade.	High
BR4	Each segment must come with a suggested offer, channel, and timing.	High
BR5	Segments must be easy for marketing and sales to reach and use.	High
BR6	The project must follow data privacy rules and handle sensitive data with care.	High
BR7	The project must explain how missing or unusual data was handled.	Medium
BR8	The project must include a plan to measure success.	Medium
BR9	The project should suggest how often segments should be reviewed.	Low

9. Constraints and Risks

Constraints

- **Privacy:** Customer data must be handled under data protection rules. Sensitive details should be used carefully.
- **Data quality:** Missing values, errors, and unusual cases (such as very heavy business users) must be handled before analysis.
- **Simplicity:** Results must be clear to people who are not data experts.
- **Usefulness:** A segment is only valuable if TelcoX can reach it with a different offer through an existing channel.

Risks

Risk	Possible effect	How to reduce it
Poor or missing data	Segments may be unreliable.	Check data early and record how problems were fixed.
Too many segments	Marketing cannot act on all of them.	Aim for a small number that are clearly different.

Risk	Possible effect	How to reduce it
Segments are hard to explain	Teams will not use them.	Give each segment a simple name and short description.
Customer behavior changes	Segments become out of date.	Plan regular reviews.
Privacy concerns	Legal and trust problems.	Avoid sensitive details and keep data protected.

10. Success Measures

Measure	What it tells us
Promotion response rate by segment	Whether targeted offers work better than broad ones.
Upgrade rate among targeted customers	Whether segments help find customers ready to upgrade.
Change in ARPU	Whether revenue per customer grows.
Marketing cost per upgrade	Whether spending is more efficient.
Team use of segments	Whether marketing and sales actually use the segments.

Target numbers are not set yet. They will be defined in document 09, the KPI and Measurement Plan.

11. Expected Deliverables

1. This Business Requirements Document
2. Stakeholder Analysis
3. Requirements and User Stories
4. Data Dictionary
5. Data Quality and Preparation Report
6. Segmentation Analysis (notebook)
7. Segment Profiles
8. Marketing Recommendations
9. KPI and Measurement Plan
10. Executive Summary Presentation

12. Approach in Brief

1. Collect and combine customer data.
2. Clean the data and create useful measures, such as average monthly data use.
3. Use K-means clustering to find groups of customers with similar behavior.
4. Describe and name each group.
5. Recommend a marketing action for each group.
6. Define how results will be measured.

K-means is a method that sorts customers into groups so that people in the same group behave alike and people in different groups behave differently. The groups come from the data itself, not from labels set in advance.

13. Approval

Name	Role	Date	Status
	Marketing lead		Pending
	Product lead		Pending
	Data team lead		Pending

Approvals are shown as placeholders to demonstrate a normal BRD process.

Stakeholder Analysis

Project name	TelcoX Customer Segmentation
Document	02 Stakeholder Analysis
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related document	01 Business Requirements Document

1. Purpose of This Document

This document looks closely at the people and teams affected by the customer segmentation project. For each one, it explains what they need, how they will use the results, what worries them, and how the project team should keep them involved.

Knowing this early helps the project deliver segments that people will actually use.

2. Stakeholder List

ID	Stakeholder	Role in the project	Main interest
S1	Marketing team	Main user of the segments	Better campaigns and higher response
S2	Product and pricing team	Uses results to review plans	Plans that match real usage
S3	Sales and retail team	Uses segments with customers face to face	Simple guidance for upgrade conversations
S4	Customer care team	Supports customers after offers are made	Fewer complaints and better service
S5	IT and data team	Provides data and systems	Clear data needs and manageable workload
S6	Leadership	Sponsors the project and approves it	Higher ARPU and a good return on spending
S7	Legal and compliance team	Checks privacy and data rules	Safe and lawful use of customer data
S8	Customers	Receive the offers	Relevant offers and respect for their privacy

Customers do not take part in the project directly. They are included because the project affects them most.

3. Stakeholder Details

S1. Marketing team

- **What they need:** Clear customer groups they can target, with simple descriptions and suggested offers.
- **How they will use the results:** To plan campaigns, choose who gets which offer, and decide where to spend the budget.
- **Main concerns:** Segments that are too complicated, too many groups to manage, or groups they cannot reach through current channels.
- **What success looks like to them:** Higher response to promotions and less money spent on offers that do not fit.

S2. Product and pricing team

- **What they need:** To see whether the current plans match how customers really use their services.
- **How they will use the results:** To spot gaps between plans and customer needs, and to consider changes to plans in the future.
- **Main concerns:** Results that suggest changes to plans that are hard to carry out.
- **What success looks like to them:** Clear evidence about which customer groups need a different plan.

S3. Sales and retail team

- **What they need:** Short, simple segment descriptions they can use in shops and calls.
- **How they will use the results:** To recognize which customers may be ready to upgrade and to offer the right plan.
- **Main concerns:** Long reports, technical language, or extra steps that slow down service.
- **What success looks like to them:** More successful upgrade conversations with less effort.

S4. Customer care team

- **What they need:** To know what problems and questions are common in each segment.
- **How they will use the results:** To prepare for questions about new offers and to improve how they help each group.
- **Main concerns:** A rise in complaints if customers receive offers that do not suit them.
- **What success looks like to them:** Fewer complaints about irrelevant messages.

S5. IT and data team

- **What they need:** A clear list of data fields, where they come from, and how often they are needed.
- **How they will use the results:** To prepare the data and, later, load segment labels into the systems that marketing uses.
- **Main concerns:** Heavy extra workload, unclear data requests, and data quality problems.
- **What success looks like to them:** A clear request, a clean process, and results that are easy to maintain.

S6. Leadership

- **What they need:** Proof that segmentation improves business results.
- **How they will use the results:** To decide whether to continue, expand, or fund the approach.
- **Main concerns:** Cost, delays, and unclear benefits.
- **What success looks like to them:** Growth in upgrades and ARPU, and a better return on marketing spending.

S7. Legal and compliance team

- **What they need:** Confidence that customer data is used lawfully and safely.
- **How they will use the results:** To review what data is used and how segments are stored and shared.
- **Main concerns:** Use of sensitive personal details, and data being shared with people who should not see it.
- **What success looks like to them:** A project that follows data protection rules with no complaints.

S8. Customers

- **What they need:** Offers that fit their needs and respect for their privacy.
- **How they will be affected:** They will receive more relevant promotions and fewer unwanted messages.
- **Main concerns:** Too many messages, offers that feel pushy, and how their data is used.
- **What success looks like to them:** Offers that are useful and easy to understand.

4. Influence and Interest

This table rates how much power each stakeholder has over the project (influence) and how much they care about the result (interest). The ratings are High, Medium, or Low.

Stakeholder	Influence	Interest	Priority
Leadership	High	High	Manage closely
Marketing team	High	High	Manage closely
Product and pricing team	Medium	High	Keep informed and involved
IT and data team	High	Medium	Keep satisfied

Stakeholder	Influence	Interest	Priority
Legal and compliance team	High	Low	Keep satisfied
Sales and retail team	Medium	Medium	Keep informed
Customer care team	Low	Medium	Keep informed
Customers	Low	High	Protect their interests

What the priorities mean

- **Manage closely:** Involve them often and ask for their input on key decisions.
- **Keep satisfied:** Make sure their needs are met, and update them at important points.
- **Keep informed:** Share results and changes regularly.
- **Protect their interests:** They are not part of the team, but decisions should consider how they are affected.

5. Where Stakeholder Needs May Conflict

Different teams want different things. Knowing this early helps avoid problems later.

Possible conflict	Teams involved	Suggested way forward
Marketing wants many detailed segments, while sales wants only a few simple ones.	Marketing and Sales	Aim for a small number of clear segments and agree on the final number together.
Marketing wants to use as much customer data as possible, while legal wants to limit it.	Marketing and Legal	Use only the data that is needed, and review the data list with legal early.
Leadership wants quick results, while IT needs time to prepare clean data.	Leadership and IT	Agree on a realistic timeline and share progress at set points.
Teams define a "high-value customer" differently.	Marketing, Sales, Product	Agree on one shared definition at the start of the project.

6. Communication Plan

Stakeholder	What to share	How	How often
Leadership	Progress summary, key findings, expected business impact	Short summary and presentation	At each major step
Marketing team	Draft segments, profiles, and offer ideas	Workshop and written profiles	Every week
Product and pricing team	Usage findings and plan gaps	Meeting and short report	At key steps
IT and data team	Data requests and data quality findings	Written request and check-in meeting	Every week at the start, then as needed
Legal and compliance team	Data list and privacy approach	Written review	Before analysis begins and before results are shared
Sales and retail team	Simple segment guide	One-page summary	When segments are final
Customer care team	Segment summary and expected questions	Short briefing	Before campaigns start

7. Stakeholder Questions to Ask

These questions will guide early conversations with each team.

Marketing

- Which campaigns run today, and which results do you track?

- How many segments can your team realistically manage?
- Which channels can you use to reach customers?

Product and pricing

- Which plans are the most and least popular?
- What do you believe stops customers from upgrading?

Sales and retail

- What do customers ask for before they upgrade?
- What information would help you most during a conversation?

Customer care

- Which complaints are linked to promotions or plan changes?

IT and data

- Which systems hold the data, and how can it be combined?
- How often is the data updated?

Leadership

- What improvement would make this project a success?
- What is the time and budget limit?

Legal and compliance

- Which customer details are allowed in this analysis?
- What rules apply to storing and sharing segment labels?

8. Summary

The most important stakeholders are leadership and the marketing team, so they will be involved most closely. IT, legal, and product teams have a strong effect on what is possible, so they will be consulted early. Sales, customer care, and customers are affected by the results, so their needs will shape how segments are described and how offers are made.

The main points to settle early are the number of segments, one shared meaning of "high-value customer," and which customer data can be used.

Requirements and User Stories

Project name	TelcoX Customer Segmentation
Document	03 Requirements and User Stories
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	01 Business Requirements, 02 Stakeholder Analysis

1. Purpose of This Document

The Business Requirements Document explains what TelcoX wants. This document explains what the project must do to deliver it. It turns each business need into specific requirements and user stories that can be checked one by one.

Each item has an ID, a priority, and a link back to the business requirement it supports.

2. How to Read This Document

Priority levels

Priority	Meaning
Must have	The project fails without it.
Should have	Important, but the project still works without it.
Could have	Useful extra, done only if time allows.

ID labels

Label	Meaning
BR	Business requirement (from document 01)
FR	Functional requirement (what the project must do)
NFR	Quality requirement (how well it must be done)
US	User story (a need written from the user's point of view)

3. Functional Requirements

3.1 Data preparation

ID	Requirement	Linked to	Priority
FR1	Combine data from billing, the customer database, network usage records, and the campaign tool, using the customer ID.	BR1, BR7	Must have
FR2	Check the data for missing values, duplicate records, and unusual values, and record what was found.	BR7	Must have
FR3	Decide and document how each data problem was handled (for example, removed, filled in, or kept).	BR7	Must have
FR4	Create customer measures from the raw data, such as average monthly data use, calls per month, top-up frequency, average top-up amount, and months as a customer.	BR1	Must have

ID	Requirement	Linked to	Priority
FR5	Put all measures on a similar scale so that no single measure has more weight than the others.	BR1	Must have

3.2 Segmentation

ID	Requirement	Linked to	Priority
FR6	Test different numbers of segments and choose the best number using the elbow method and silhouette score. Explain the choice in simple words.	BR1	Must have
FR7	Run K-means clustering and place every customer in exactly one segment.	BR1	Must have
FR8	Run the model again with different starting points to check that the segments stay mostly the same.	BR1	Should have

3.3 Segment profiles and upgrade potential

ID	Requirement	Linked to	Priority
FR9	Describe each segment by size, typical usage, typical spending, and main demographics.	BR2	Must have
FR10	Give each segment a short, plain name and a description of two to three sentences.	BR2	Must have
FR11	Estimate the upgrade potential of each segment, using signs such as usage above the current plan limit and past upgrade behavior. Rank segments from highest to lowest potential.	BR3	Must have

3.4 Marketing actions

ID	Requirement	Linked to	Priority
FR12	Recommend an offer, a channel, and a timing for each segment.	BR4	Must have
FR13	Confirm that each recommended channel is one TelcoX already uses.	BR5	Must have
FR14	Provide a one-page guide for each segment that sales and retail staff can use.	BR5	Should have

3.5 Privacy

ID	Requirement	Linked to	Priority
FR15	Remove direct personal details (such as names and phone numbers) from the analysis data.	BR6	Must have
FR16	Use only the data fields that are needed, and list why each one is needed.	BR6	Must have

3.6 Measurement and reporting

ID	Requirement	Linked to	Priority
FR17	Define the measures of success (response rate, upgrade rate, ARPU change, cost per upgrade) and how each one is calculated.	BR8	Must have
FR18	Describe a test that compares targeted customers with a control group that receives the usual offer.	BR8	Must have
FR19	Present the findings in a short summary deck for non-technical readers.	BR2, BR8	Must have
FR20	Recommend how often segments should be refreshed.	BR9	Could have
FR21	Build a simple dashboard that shows segments with filters.	BR5	Could have

4. Quality Requirements

ID	Requirement	How it will be checked	Priority
NFR1	Plain language. All documents and segment descriptions must be understandable to a reader with no data background.	A non-technical reader can explain each segment in one sentence after reading.	Must have
NFR2	Repeatable results. The notebook must give the same results each time it is run.	Run the notebook from start to finish twice and compare the segment sizes.	Must have
NFR3	Clear segments. The number of segments must be small enough for marketing to act on.	The final model has between 4 and 6 segments.	Must have
NFR4	Meaningful segment size. No segment should be too small to target.	Each segment holds at least 5 percent of customers.	Should have
NFR5	Documented data. Every field used must be described in the data dictionary.	Each field in the notebook appears in document 04.	Must have
NFR6	Readable code. The notebook must have short explanations before each step.	Each section starts with a plain description of what it does.	Should have

5. User Stories

Each story follows this format: *As a [role], I want [need], so that [benefit]*. The acceptance criteria are the checks that show the story is done.

Epic A: Data preparation

US1. Combined customer data | Must have | Linked to FR1, FR4, FR5

As an IT and data analyst, I want customer data from all sources joined into one table, so that every customer can be analyzed from a single view.

Acceptance criteria:

- ☐ One row exists for each customer.
- ☐ Billing, usage, transaction, and demographic fields all appear in the table.
- ☐ The number of customers before and after joining is recorded.

US2. Data quality findings | Must have | Linked to FR2, FR3

As a data team lead, I want a clear record of data problems and how they were fixed, so that I can trust the results.

Acceptance criteria:

- ☐ Missing values are counted for each field.
- ☐ Duplicate records and unusual values are listed.
- ☐ Each fix is explained in one sentence.

Epic B: Segmentation

US3. Clear customer groups | Must have | Linked to FR6, FR7

As a marketing manager, I want customers placed into a small number of distinct groups, so that I can plan different campaigns for each group.

Acceptance criteria:

- ☐ Every customer belongs to exactly one segment.
- ☐ The final model has between 4 and 6 segments.
- ☐ The reason for the chosen number is explained in simple words.

US4. Trustworthy segments | Should have | Linked to FR8, NFR2

As a product manager, I want to know the segments are stable, so that I do not base plans on groups that change each time the analysis is run.

Acceptance criteria:

- ☐ The model is run at least three times with different starting points.
- ☐ Segment sizes are compared across the runs.
- ☐ Any large differences are explained.

Epic C: Segment profiles

US5. Easy-to-understand segments | Must have | Linked to FR9, FR10

As a sales and retail agent, I want each segment to have a short name and description, so that I can quickly recognize which customers may suit an upgrade.

Acceptance criteria:

- ☐ Each segment has a name of five words or fewer.
- ☐ Each segment has a description of two to three sentences.
- ☐ Descriptions use everyday words, with no technical terms.

US6. Segment detail for marketing | Must have | Linked to FR9

As a marketing manager, I want to see the size, usage, spending, and demographics of each segment, so that I can understand who these customers are.

Acceptance criteria:

- ☐ A profile table shows each segment side by side.
- ☐ Each profile shows the share of total customers.
- ☐ Each profile shows at least three usage or spending measures.

Epic D: Upgrade potential

US7. Upgrade ranking | Must have | Linked to FR11

As a leadership member, I want segments ranked by how ready they are to upgrade, so that budget goes to the groups most likely to respond.

Acceptance criteria:

- ☐ Segments are ranked from highest to lowest upgrade potential.
- ☐ The signs used to judge potential are listed.
- ☐ The top-ranked segments are named in the summary deck.

US8. Plan fit | Should have | Linked to FR11

As a product manager, I want to see which segments use more than their current plan allows, so that I can see where plans do not match real needs.

Acceptance criteria:

- ☐ For each segment, the share of customers above their plan limits is shown.
- ☐ The segments with the largest gap are highlighted.

Epic E: Marketing actions

US9. Tailored offers | Must have | Linked to FR12, FR13

As a marketing manager, I want a suggested offer, channel, and timing for each segment, so that I can launch campaigns without starting from scratch.

Acceptance criteria:

- ☐ Every segment has one offer, one channel, and one timing.
- ☐ Each recommendation explains in one sentence why it suits the segment.
- ☐ Every channel is one that TelcoX already uses.

US10. Retail guide | Should have | Linked to FR14

As a retail agent, I want a one-page guide for each segment, so that I can make the right offer during a customer visit.

Acceptance criteria:

- ☐ Each guide fits on one page.
- ☐ Each guide includes the segment description, a suggested offer, and one example conversation starter.

US11. Customer care readiness | Could have | Linked to FR12

As a customer care agent, I want to know what questions each segment is likely to ask about an offer, so that I can answer quickly.

Acceptance criteria:

- ☐ Each segment has at least two likely questions listed.

Epic F: Privacy

US12. Safe use of data | Must have | Linked to FR15, FR16

As a legal and compliance officer, I want only the needed data used and personal details removed, so that customer privacy is protected.

Acceptance criteria:

- ☐ The analysis data contains no names, phone numbers, or addresses.
- ☐ Every data field used has a written reason.
- ☐ Sensitive fields are either left out or marked as sensitive with a reason for use.

Epic G: Measurement

US13. Measures of success | Must have | Linked to FR17

As a leadership member, I want clear measures of success, so that I can judge whether segmentation is worth continuing.

Acceptance criteria:

- ☐ Response rate, upgrade rate, ARPU change, and cost per upgrade are each defined.
- ☐ Each definition includes how it is calculated.
- ☐ Each measure is compared against the current broad approach.

US14. Fair test | Must have | Linked to FR18

As a marketing manager, I want a test that compares targeted offers with the usual offers, so that I can prove the new approach works.

Acceptance criteria:

- ☐ The test has a targeted group and a control group.
- ☐ Group selection is described in simple steps.
- ☐ The test duration and the decision rule are stated.

US15. Executive summary | Must have | Linked to FR19

As a leadership member, I want a short summary of the problem, findings, and recommendations, so that I can make a decision in a few minutes.

Acceptance criteria:

- ☐ The deck is ten slides or fewer.
- ☐ It covers the problem, the segments, the ranking, the recommendations, and the next steps.
- ☐ It contains no technical terms without a short explanation.

US16. Refresh plan | Could have | Linked to FR20

As a data team lead, I want to know how often segments should be updated, so that they stay accurate as customer behavior changes.

Acceptance criteria:

- ☐ A refresh schedule is recommended with a reason.
- ☐ One sign that the segments are out of date is described.

6. Traceability

This table shows that every business requirement is covered by at least one functional requirement and one user story.

Business requirement	Functional requirements	User stories
BR1 Group customers by behavior	FR1, FR4 to FR8	US1, US3, US4
BR2 Clear names and descriptions	FR9, FR10, FR19	US5, US6, US15
BR3 Show upgrade potential	FR11	US7, US8
BR4 Offer, channel, and timing	FR12	US9, US11
BR5 Easy for teams to use	FR13, FR14, FR21	US9, US10
BR6 Data privacy	FR15, FR16	US12
BR7 Explain data handling	FR1 to FR3	US2
BR8 Measurement plan	FR17 to FR19	US13, US14, US15
BR9 Review schedule	FR20	US16

7. Project Completion Checklist

The project is complete when:

- ☐ All Must have requirements are met.
- ☐ All acceptance criteria for Must have user stories are checked off.
- ☐ Every business requirement appears in the traceability table.
- ☐ The notebook runs from start to finish and gives the same segments each time.
- ☐ The summary deck has been read by a non-technical person who could explain the segments back.

Data Dictionary

Project name	TelcoX Customer Segmentation
Document	04 Data Dictionary
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	01 Business Requirements, 03 Requirements and User Stories

1. Purpose of This Document

This document lists every data field used in the project. For each field it explains what the field means, what values to expect, where the data comes from, and how it will be used. Anyone reading the analysis can use this document to understand the data without asking questions.

It supports requirement FR16 (list why each field is needed) and quality requirement NFR5 (every field is documented).

2. About the Dataset

Item	Description
Dataset name	TelcoX customer table
Type	Synthetic (created by the script <code>src/generate_telcox_data.py</code> with a fixed random seed of 42, not real customer data)
Size	15,000 customers (15,060 rows, because a few records are repeated on purpose)
One row means	One customer
Time period	A snapshot at the end of one month. Usage and spending are averages of the last three months.
Currency	All amounts are in "currency units" so the case study is not tied to one country.
Source systems	Customer database (CRM), billing, network usage records, campaign tool

All customers are linked by one customer ID. The ID is made up and does not point to any real person.

3. Plan Tiers

Each customer is on one of four plans. These are used to work out whether a customer is using more than their plan allows.

Plan	Monthly price	Data allowance (GB)	Voice allowance (minutes)
Basic	10	5	100
Standard	20	15	300
Plus	35	40	600
Premium	55	100	1,000

4. Field List

4.1 Identifier

Field	Meaning	Type	Example values	Source	Used for
customer_id	A made-up code that identifies one customer.	Text	TX000001	CRM	Joining tables only

Field	Meaning	Type	Example values	Source	Used for
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4.2 Demographic fields

Field	Meaning	Type	Example values	Source	Used for	Sensitive
age_group	The customer's age band.	Category	18-24, 25-34, 35-44, 45-54, 55+	CRM	Profiling	No
gender	The customer's gender as recorded.	Category	Female, Male, Not stated	CRM	Fairness check only	Yes
area_type	Whether the customer lives in a city or a rural area.	Category	Urban, Rural	CRM	Profiling	No
region	The part of the country where the customer lives.	Category	North, South, East, West, Central	CRM	Profiling	No
registration_channel	How the customer first signed up.	Category	Retail shop, Mobile app, Self-service, Reseller	CRM	Profiling	No

4.3 Account fields

Field	Meaning	Type	Example values	Source	Used for	Sensitive
account_type	Whether the customer pays before use or after use.	Category	Prepaid, Postpaid	CRM	Segmentation	No
tenure_months	Number of months as a TelcoX customer.	Whole number	1 to 120	CRM	Segmentation	No
current_plan	The plan the customer is on now.	Category	Basic, Standard, Plus, Premium	Billing	Segmentation	No
plan_price	Monthly price of the current plan.	Number	10, 20, 35, 55	Billing	Working out ratios	No
plan_data_gb	Monthly data allowance of the current plan.	Number	5, 15, 40, 100	Billing	Working out ratios	No
plan_voice_min	Monthly voice allowance of the current plan.	Whole number	100, 300, 600, 1000	Billing	Working out ratios	No
has_home_broadband	Whether the customer also has a home broadband service.	Yes or No	Yes, No	Billing	Segmentation	No

4.4 Usage fields

All usage fields are monthly averages over the last three months.

Field	Meaning	Type	Example values	Source	Used for
avg_data_gb	Mobile data used per month, in gigabytes.	Number	0.2 to 120	Network records	Segmentation
avg_voice_min	Call minutes used per month.	Number	0 to 1,500	Network records	Segmentation
avg_sms	Text messages sent per month.	Whole number	0 to 500	Network records	Segmentation
offpeak_share	Share of usage that happens at off-peak times (night and weekends). 0 means none, 1 means all.	Number	0.00 to 1.00	Network records	Segmentation
roaming_days_3m	Number of days the customer used the service abroad in the last three months.	Whole number	0 to 90	Network records	Segmentation
broadband_gb	Home broadband data used per month. Empty if the customer has no broadband.	Number	20 to 800	Network records	Segmentation
months_over_data_limit	Number of the last three months in which data use went above the plan allowance.	Whole number	0 to 3	Network records	Segmentation

4.5 Transaction fields

Field	Meaning	Type	Example values	Source	Used for
avg_monthly_spend	Average amount the customer pays per month.	Number	2 to 120	Billing	Segmentation
topups_per_month	How many times the customer tops up each month. Empty for postpaid customers.	Number	0.5 to 12	Billing	Segmentation
avg_topup_amount	Average value of one top-up. Empty for postpaid customers.	Number	1 to 50	Billing	Segmentation
bundle_purchases_3m	Number of extra bundles bought in the last three months.	Whole number	0 to 15	Billing	Segmentation
late_payments_6m	Number of late bill payments in the last six months. Empty for prepaid customers.	Whole number	0 to 6	Billing	Profiling

4.6 Plan change and campaign fields

Field	Meaning	Type	Example values	Source	Used for
upgrades_24m	Number of times the customer moved to a higher plan in the last 24 months.	Whole number	0 to 3	CRM	Upgrade potential
months_since_last_upgrade	Months since the last upgrade. Empty if the customer never upgraded.	Whole number	1 to 24	CRM	Upgrade potential
promos_received_6m	Number of promotions sent to the customer in the last six months.	Whole number	0 to 12	Campaign tool	Segmentation
promos_accepted_6m	Number of those promotions the customer accepted.	Whole number	0 to 12	Campaign tool	Segmentation

5. Measures Created From the Fields

Some of the most useful measures are not in the raw data. They are worked out from the fields above during data preparation (requirement FR4).

Measure	How it is worked out	What it tells us
data_use_ratio	avg_data_gb divided by plan_data_gb	How much of the data allowance the customer uses. A value above 1 means the customer goes over the limit.
voice_use_ratio	avg_voice_min divided by plan_voice_min	How much of the voice allowance the customer uses.
topup_value_per_month	topups_per_month multiplied by avg_topup_amount	Total top-up spending per month for prepaid customers.
promo_response_rate	promos_accepted_6m divided by promos_received_6m	How often the customer says yes to offers. Empty if no promotions were received.
has_upgraded	1 if upgrades_24m is above 0, otherwise 0	Whether the customer has upgraded before.

6. Which Fields Are Used Where

To keep the segments based on behavior, the analysis separates fields into four groups.

Group	Fields	Role
Used to create segments	Usage fields, transaction fields (except late payments), account type, tenure, home broadband use (broadband_gb, with 0 for no broadband), the measures in section 5, promos received, and promo response rate	These decide which customers are grouped together.

Group	Fields	Role
Used to describe segments	Age group, area type, region, registration channel, current plan, late payments	Checked after the segments are made, to explain who is in each group.
Used for upgrade potential	data_use_ratio, voice_use_ratio, months_over_data_limit, upgrades_24m, months_since_last_upgrade, promo_response_rate	Combined to rank segments by how ready they are to upgrade.
Not used in the analysis	customer_id (only for joining), gender (only for a fairness check after segments are made)	Kept out so segments do not depend on personal traits.

7. Expected Data Quality Issues

The dataset is built to include the kinds of problems found in real data, so that the preparation work in requirement FR2 is meaningful.

Field or issue	What to expect	Planned handling
age_group	Missing for a share of prepaid customers.	Keep as a "Not stated" group, because removing these customers would lose a large part of the prepaid base.
gender	Some records marked "Not stated".	Keep as a separate value.
topups_per_month, avg_topup_amount	Empty for postpaid customers by design.	Not an error. Handle by account type.
late_payments_6m	Empty for prepaid customers by design.	Not an error. Handle by account type.
broadband_gb	Empty for customers without broadband by design.	Fill with 0 for analysis.
avg_data_gb, avg_voice_min	A small number of very high values (heavy business users).	Review them, then decide whether to keep, cap, or analyze separately.
Duplicate rows	A few repeated customer records.	Remove duplicates and record how many were removed.
Inconsistent values	A few records where the plan allowance does not match the plan name.	Correct using the plan table in section 3.

8. Privacy Notes

- The dataset contains no names, phone numbers, addresses, or national ID numbers.
- Every customer ID is made up.
- Gender is the only sensitive field. It is not used to create segments.
- Because the data is synthetic, no real person can be identified from it. In a real project, the same fields would still need approval from the legal and compliance team (see US12).

Data Quality and Preparation Report

Project name	TelcoX Customer Segmentation
Document	05 Data Quality and Preparation Report
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	03 Requirements and User Stories, 04 Data Dictionary, Segmentation Analysis notebook (Stage 1)

1. Purpose of This Document

This report explains what was wrong with the TelcoX customer data, how each problem was handled, and how the data was prepared for grouping customers. It supports requirements FR2 to FR5 and user story US2. Every number in this report comes from Stage 1 of the segmentation notebook, so anyone can rerun the notebook and check it.

2. Summary

Item	Result
Rows received	15,060
Customers after cleaning	15,000
Repeated customer records removed	60 (0.4 percent)
Records with a wrong plan allowance, corrected	33
Customers with a missing age	1,649 (11.0 percent)
Customers with gender marked "Not stated"	616 (4.1 percent)
Fields that are empty by design	5
Customers who received no promotions	91
Customers with at least one extreme value limited	1,484 (10 percent)
Features considered for grouping	21
Features used for grouping	19

In short: the data was in good shape. The problems were small and easy to fix, apart from the missing ages, which are mostly in prepaid customers. No customers were removed except exact repeats.

3. Joining the Data Sources

The data dictionary lists four source systems: billing, the customer database, network usage records, and the campaign tool. In a real project, these would be joined using the customer ID, and the number of customers before and after joining would be recorded.

In this case study the data is synthetic, and the generator already produces one table with fields from all four sources. So no join was needed. The one check that applies is the row count: the table had 15,060 rows and 15,000 customers once repeated records were removed.

4. Problems Found and How They Were Handled

4.1 Overview

#	Problem	How many	How it was handled	Reason
1	Repeated customer records	60	Kept the first record, removed the rest	Each customer should count once.
2	Plan allowance does not match the plan name	33	Replaced with the allowance from the plan table	The plan name is the more reliable field.
3	Age not recorded	1,649	Labelled "Not stated"	Removing them would lose a large part of the prepaid base.
4	Broadband data empty (no broadband)	11,711	Set to 0	These customers use no home broadband data.
5	Top-up fields empty (postpaid)	7,146	Set to 0	Postpaid customers do not top up.
6	Promo response rate empty (no promotions received)	91	Left empty in the clean table. Set to 0 only in the copy used for grouping.	A response rate cannot be worked out without promotions.
7	Very large values in usage and spending	1,484 customers	Limited in the grouping copy only (see section 5.3)	A few very heavy users would otherwise pull the groups towards themselves.

4.2 Missing values

Six fields contain empty cells. Five of them are empty for a good reason. Only the age field is a real gap.

Field	Empty cells	Share of customers	Reason	Action
months_since_last_up_grade	12,354	82.4%	Customer never upgraded	Left empty. Used only to describe segments.
broadband_gb	11,711	78.1%	No home broadband	Set to 0
late_payments_6m	7,854	52.4%	Prepaid customers have no bills	Left empty. Used only to describe segments.
topups_per_month	7,146	47.6%	Postpaid customers do not top up	Set to 0
avg_topup_amount	7,146	47.6%	Postpaid customers do not top up	Set to 0
age_group	1,649	11.0%	Age was not recorded	Labelled "Not stated"

About the missing ages. The gap is not even across customer types. Age is missing for 18.3 percent of prepaid customers but only 3.0 percent of postpaid customers. This matches how real telecom data often looks, because prepaid customers can sign up with less information. Age is not used to build the segments, so the gap does not affect how customers are grouped. It only affects how well the age of each segment can be described, and the "Not stated" group is shown openly in the profiles.

About gender. 616 customers (4.1 percent) have gender marked "Not stated". This is stored as a value, not as an empty cell. Gender is used only for a fairness check after the segments are made.

4.3 Repeated records and wrong plan records

- **Repeated records.** 60 customer IDs appeared twice. The extra copies were removed.
- **Wrong plan allowances.** In 33 records, the data or call allowance did not match the plan name. For example, a Basic plan customer showing the allowance of a Plus plan. All three plan fields (price, data allowance, call allowance) were checked against the plan table, and the plan name was treated as correct.

4.4 Extreme values

Usage is heavily skewed. Most customers use little, and a small group uses many times more than a typical customer.

Field	Typical customer (median)	99th percentile	Largest value
Mobile data per month (GB)	6.9	54.3	119.4
Call minutes per month	167.7	1,190.7	1,500.0
Texts per month	64.0	298.4	500.0
Home broadband data (GB)	107.2	336.9	777.3

These figures are from the file before cleaning. The largest values are far from the typical customer, for example 119 GB of mobile data against a median of 7 GB. They are plausible, such as business users, so the rows were kept. The limit was applied only to the copy of the data used for grouping.

5. How the Data Was Prepared for Grouping

5.1 New measures

Five measures were worked out from existing fields (requirement FR4), plus two Yes or No fields turned into 1 or 0.

Measure	How it is worked out
data_use_ratio	Mobile data used divided by the data allowance
voice_use_ratio	Call minutes used divided by the call allowance
topup_value_per_month	Top-ups per month multiplied by average top-up amount
promo_response_rate	Promotions accepted divided by promotions received
has_upgraded	1 if the customer upgraded in the last 24 months, otherwise 0
is_prepaid	1 if the account is prepaid, otherwise 0
has_broadband	1 if the customer has home broadband, otherwise 0

5.2 Choosing the features

Only behavior fields were used to build the segments. Age, area, region, sign-up channel, current plan, and gender were left out, so the groups come from behavior. They are used afterwards to describe the segments. This gave 21 candidate features.

Removing repeated information. If two features move together almost perfectly, the grouping method counts the same information twice. Three pairs had a correlation above 0.8.

Pair	Correlation	Decision	Reason
broadband_gb and has_broadband	0.88	Dropped has_broadband	Broadband use is 0 when there is no broadband, so it already shows who has it.
promo_response_rate and promos_accepted_6m	0.85	Dropped promos_accepted_6m	The response rate already contains the number accepted.
is_prepaid and topup_value_per_month	0.81	Kept both	Top-up value is 0 for every postpaid customer, which explains the link. Among prepaid customers it also shows spending size.

The final list has 19 features.

5.3 Limiting extreme values

For 12 heavy-tailed fields (usage, spending, top-ups, bundles, roaming, broadband, and the two use ratios), any value above the 99th percentile was set to the 99th percentile value. This was done on a separate copy, so the clean table and the segment profiles keep the real values.

- Between 103 and 150 values were limited in each of the 12 fields.
- In total, 1,484 customers (10 percent) had at least one value limited. Almost all of them had only one or two.

5.4 Scaling

Call minutes run into the hundreds, while a Yes or No flag is only 0 or 1. The grouping method measures distance, so large numbers would dominate. Every feature was scaled to an average of 0 and a spread of 1 (requirement FR5). The notebook checks this by printing the mean and spread of every scaled feature.

6. Privacy Check

- The dataset contains no names, phone numbers, addresses, or national ID numbers (requirement FR15).
- Customer IDs are made up.
- Every field used has a written purpose in the data dictionary (requirement FR16).
- Gender is the only sensitive field. It is kept out of the segmentation and used only for a fairness check.

7. Limits of This Review

- **The data is synthetic.** The problems were added on purpose by the data generator, so they are tidier than real problems. Real data would need more checks, such as impossible dates, customer IDs that do not match between systems, and values entered in different units.
- **Some choices affect the results.** The main ones are limiting values at the 99th percentile, setting empty broadband and top-up fields to 0, and treating missing ages as a "Not stated" group. They are reasonable and explained, but other choices would give slightly different segments.
- **No test of other choices.** We did not test how much the segments change if these choices are made differently. This could be a future improvement.

8. Requirements Covered

Requirement	Status
FR1 Combine the data sources	Not needed, because the synthetic table is already combined (section 3)
FR2 Check data quality	Done (section 4)
FR3 Document how each problem was handled	Done (section 4)
FR4 Create customer measures	Done (section 5.1)
FR5 Put measures on a similar scale	Done (section 5.4)
FR15 Remove direct personal details	Done (section 6)
FR16 Give a reason for each field used	Done (section 6 and document 04)

Segmentation Analysis Summary

Project name	TelcoX Customer Segmentation
Document	06 Segmentation Analysis Summary
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	03 Requirements and User Stories, 05 Data Quality and Preparation Report, Segmentation Analysis notebook

1. Purpose of This Document

The full analysis is in the notebook `notebooks/segmentation_analysis.ipynb`, which runs from start to finish and gives the same results each time. This document is a short, plain-language summary of what the notebook did and found, for readers who do not want to open the code. It supports requirements FR6 to FR11.

The notebook has four stages:

Stage	What happens	Requirements covered
1. Data preparation	Check the data, fix problems, create new measures, and scale everything	FR2 to FR5
2. Choose the number of segments	Test 2 to 10 groups and pick the best number	FR6, FR8
3. Build the segments	Run K-means and test stability on subsamples	FR7
4. Profile the segments	Describe, name, and rank the segments	FR9 to FR11

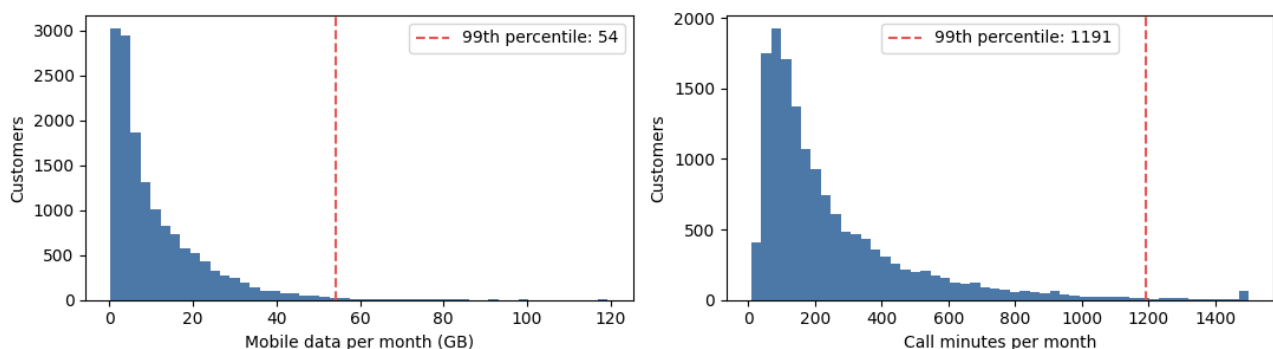
2. Stage 1: Data Preparation

Stage 1 is described in full in document 05. The main points:

- 15,060 rows became 15,000 customers after 60 repeated records were removed.
- 33 plan records with the wrong allowance were corrected.
- Five new measures were created, such as the share of the data allowance a customer uses.
- 19 behavior features were chosen. Age, area, region, sign-up channel, plan, and gender were kept out so the groups come from behavior.
- Extreme values were limited at the 99th percentile for grouping, and every feature was scaled.

The figure below shows why this mattered. Most customers use little, and a small group uses a lot.

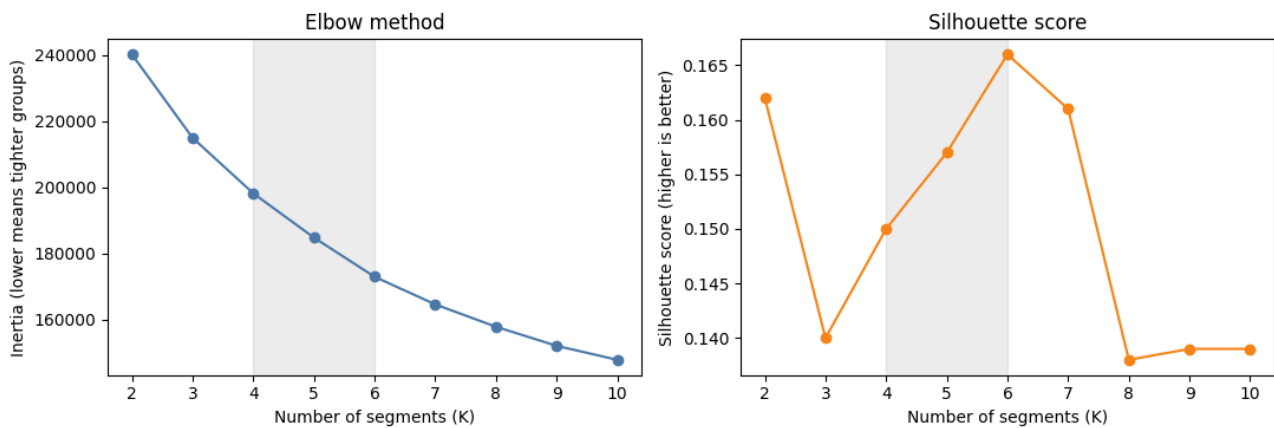
Most customers use little, and a small group uses a lot



3. Stage 2: Choosing the Number of Segments

K-means needs the number of groups in advance. We tested 2 to 10 groups with three checks: the elbow method, the silhouette score, and stability.

Groups (K)	Inertia drop from previous K	Silhouette score	Smallest group
2	not applicable	0.162	48.0%
3	10.5%	0.140	25.9%
4	7.8%	0.150	13.0%
5	6.7%	0.157	8.1%
6	6.4%	0.166	4.9%
7	4.8%	0.161	4.7%
8	4.1%	0.138	4.7%
9	3.6%	0.139	4.6%
10	2.8%	0.139	3.9%



What the evidence shows

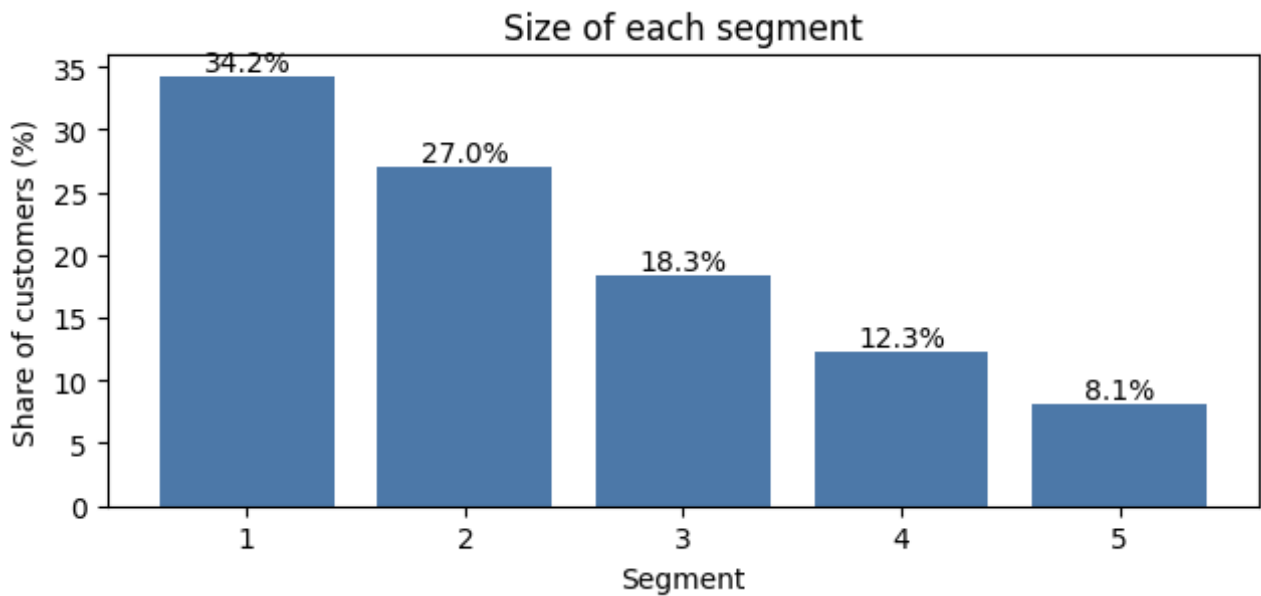
- The elbow curve has no sharp bend. The gain from each extra group shrinks gradually.
- Silhouette scores are low for every K, between 0.14 and 0.17. Customer behavior changes gradually, so the groups overlap. The segments are a practical way to divide customers, not natural separate groups.
- Two groups score well but are too coarse for marketing. The requirements ask for 4 to 6 segments.
- Within 4 to 6, K=6 scores highest, but its smallest group (4.9%) falls just under the 5 percent minimum.
- Stability is almost perfect for 4, 5, and 6, so it does not help choose between them.

Decision: K = 5. It meets every requirement and is within 0.01 of the best silhouette score in the allowed range.

4. Stage 3: Building the Segments

K-means was run with five groups, and every customer was placed in exactly one. The segments are numbered from largest to smallest.

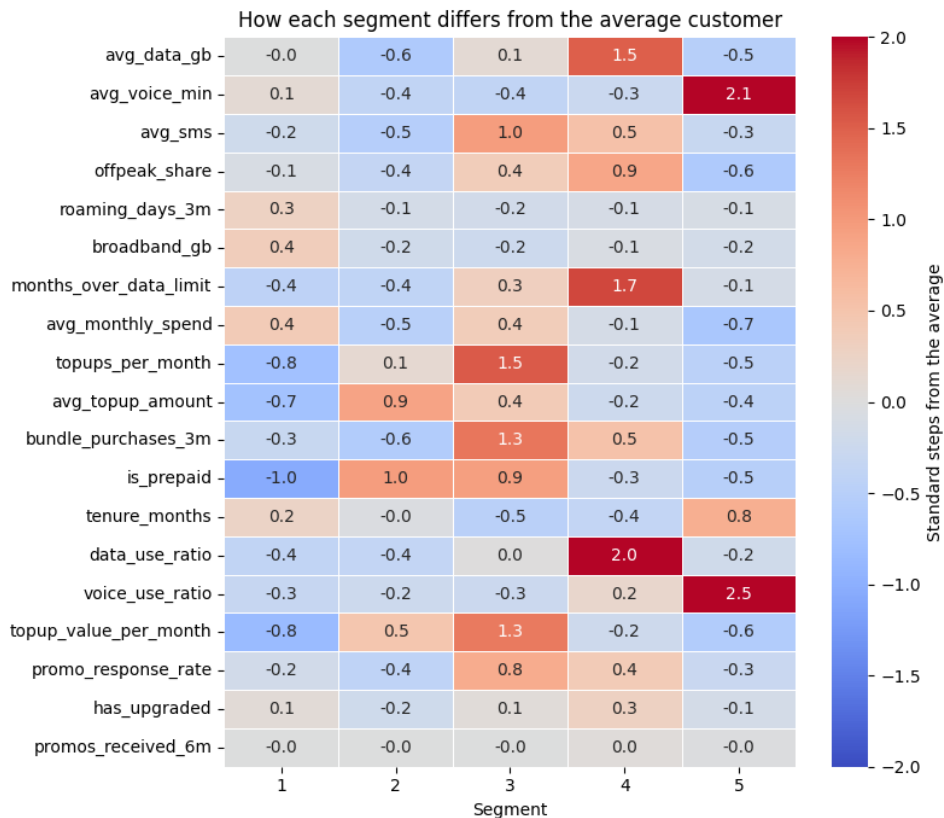
Segment	Customers	Share
1	5,137	34.2%
2	4,051	27.0%
3	2,749	18.3%
4	1,848	12.3%
5	1,215	8.1%



Stability test. We took 20 random samples of 80 percent of the customers, built new segments from each sample, and compared them with the final segments. The average adjusted Rand index was 0.993, and the lowest was 0.985. On average 99.7 percent of customers landed in the same segment, and in the worst test 99.3 percent did. The borders between segments fall in nearly the same place each time, even though the groups overlap.

5. Stage 4: Profiling and Naming the Segments

The heatmap shows how far each segment's average sits from the average of all customers, in standard steps. Red is higher than average, and blue is lower.



The five segments were named from what the data shows. Full profiles are in document 07.

Segment	Name
1	Steady Postpaid Households
2	Light Prepaid Users
3	Deal-Seeking Prepaid Users
4	Heavy Data Users
5	Long-Term Voice Users

6. Upgrade Potential

Segments were ranked using three signs. **Need** is the share of customers who use more data or calls than their plan allows and are not on the top plan. **Past behavior** is the share who upgraded in the last 24 months.

Responsiveness is the average share of promotions accepted. The score counts need for half and the other two signs for a quarter each.

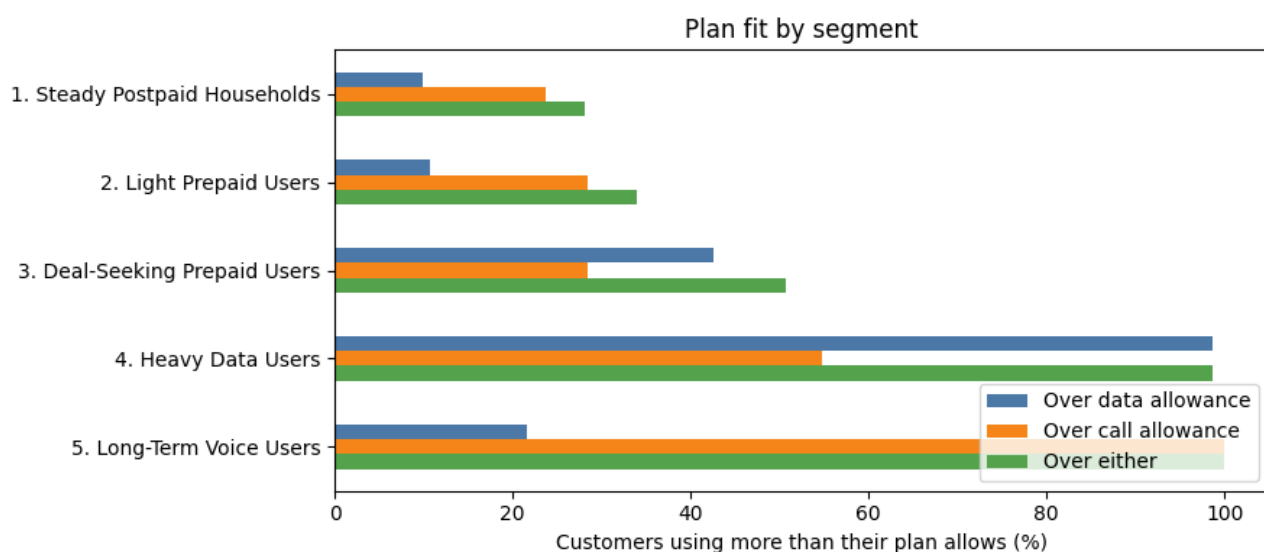
Rank	Segment	Need	Upgraded before	Promo response	Score
1	Heavy Data Users	98.6%	28.4%	20.0%	91.3
2	Long-Term Voice Users	99.3%	12.2%	7.1%	56.1
3	Deal-Seeking Prepaid Users	50.6%	19.7%	27.4%	54.6
4	Steady Postpaid Households	27.6%	20.5%	9.2%	19.3
5	Light Prepaid Users	33.7%	9.4%	4.9%	4.2

Does the ranking depend on the weights? The weights are a judgment call, so we checked two others.

Segment	Chosen weights (50 / 25 / 25)	Equal weights	Need only
Heavy Data Users	1	1	2
Long-Term Voice Users	2	3	1
Deal-Seeking Prepaid Users	3	2	3
Steady Postpaid Households	4	4	5
Light Prepaid Users	5	5	4

Heavy Data Users rank first under two of the three weightings. Long-Term Voice Users and Deal-Seeking Prepaid Users swap places 2 and 3. Steady Postpaid Households and Light Prepaid Users are always in the bottom two.

Plan fit. The chart shows the share of customers in each segment who use more than their plan allows.



7. Would Six Segments Be Better?

We built a six-segment version and compared it with the five-segment one. Five of the six groups were almost unchanged, each at least 98.7 percent made up of one original segment. The sixth group held 4.9 percent of customers and was carved out of Steady Postpaid Households, with about 12 percent of that segment moving into it. Its customers spend about 45 instead of 31, and roam abroad on about 8.5 days in three months. That is a real group, but at 4.9 percent it is just under the 5 percent minimum. We kept five segments and noted the roaming customers as a targeted add-on in document 08.

8. Limits of the Analysis

- The data is synthetic, so the segments show how the method works and do not describe real customers.
- The segments overlap, so some customers near the borders could fit two segments.
- The upgrade ranking depends on the chosen weights.
- Cleaning choices, such as limiting values at the 99th percentile, affect the segments. We did not test how much.
- Names and numbers belong to the data created with seed 42. A different seed can change them.

Segment Profiles

Project name	TelcoX Customer Segmentation
Document	07 Segment Profiles
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	03 Requirements and User Stories, 04 Data Dictionary, Segmentation Analysis notebook (Stage 4)

1. Purpose of This Document

This document describes the five customer segments found in the TelcoX customer data. It is written for marketing, sales, product, and customer care teams, so it uses plain language and avoids technical terms. It supports requirements FR9 and FR10 and user stories US5 and US6.

Each segment has its own page with a name, a short description, key facts, and a view of its upgrade potential. Ideas for offers, channels, and timing are in document 08, Marketing Recommendations.

How the segments were made. Customers were grouped by behavior only, meaning how they use TelcoX services and how they pay. Age, area, region, sign-up channel, and plan were not used to build the groups. They are used here to describe who is in each group.

Numbering. Segments are numbered from largest (Segment 1) to smallest (Segment 5).

2. The Five Segments at a Glance

#	Name	Share of customers	Account type	In one line	Upgrade rank
1	Steady Postpaid Households	34.2%	All postpaid	Long-standing customers with moderate use who mostly stay within their plans	4 of 5
2	Light Prepaid Users	27.0%	All prepaid	Low use and low spending, mostly on the Basic plan	5 of 5
3	Deal-Seeking Prepaid Users	18.3%	Almost all prepaid	Frequent small top-ups, many bundles, and the best response to promotions	3 of 5
4	Heavy Data Users	12.3%	Mixed (60% postpaid)	The heaviest data users, nearly all over their data allowance	1 of 5
5	Long-Term Voice Users	8.1%	Mostly postpaid (73%)	Very heavy callers who have been customers the longest	2 of 5

3. Side-by-Side Comparison

Measure	Seg. 1	Seg. 2	Seg. 3	Seg. 4	Seg. 5	All customers
Customers	5,137	4,051	2,749	1,848	1,215	15,000
Mobile data per month (GB)	11.1	4.6	12.3	28.8	5.5	11.3
Call minutes per month	275	162	161	186	757	252
Texts per month	69	51	137	112	63	81
Average monthly spend	32.4	19.9	31.9	25.0	17.7	26.8
Bundles bought in 3 months	1.3	0.7	5.0	3.1	0.8	2.0
Years as a customer	3.5	2.8	1.8	2.1	4.7	2.9
Prepaid	0%	100%	99.7%	39.6%	27.2%	52.4%
Has home broadband	38.4%	12.3%	10.3%	19.4%	14.4%	21.9%
On Basic plan	18%	53%	37%	65%	59%	40%

Measure	Seg. 1	Seg. 2	Seg. 3	Seg. 4	Seg. 5	All customers
Over data allowance	9.8%	10.8%	42.5%	98.6%	21.6%	28.0%
Over call allowance	23.7%	28.4%	28.4%	54.7%	100%	35.8%
Accepts promotions (average)	9.2%	4.9%	27.4%	20.0%	7.1%	12.5%
Upgraded in last 24 months	20.5%	9.4%	19.7%	28.4%	12.2%	17.6%
Aged 18 to 34	32%	31%	50%	60%	16%	37%
Aged 55 and over	17%	17%	5%	6%	33%	15%
Lives in an urban area	60%	52%	58%	72%	50%	58%
Upgrade score (0 to 100)	19.3	4.2	54.6	91.3	56.1	not applicable

Spending is in currency units per month. Age shares include customers whose age is not stated, so they add up to less than 100 percent.

4. Segment Pages

Segment 1: Steady Postpaid Households

34.2% of customers (5,137)

Summary. The largest segment. These are postpaid customers who have been with TelcoX for about three and a half years, use moderate amounts of data and calls, and mostly stay within their plans.

Key fact	This segment	All customers
Average monthly spend	32.4	26.8
Mobile data per month	11.1 GB	11.3 GB
Years as a customer	3.5	2.9
Has home broadband	38.4%	21.9%
On Plus or Premium plan	50%	25%
Customers over their data allowance	9.8%	28.0%

Who they are

- Every customer in this segment is postpaid.
- They are spread across age groups much like all customers: 32% are aged 18 to 34 and 17% are 55 and over.
- 60% live in urban areas.
- Half are on a Plus or Premium plan, the highest share of any segment. Only 18% are on Basic.
- They mostly signed up in retail shops (42%).

How they use TelcoX

- Moderate data (median 8.6 GB) and about 275 call minutes a month.
- They spend the most of any segment, with Segment 3 close behind.
- They are the most likely to have home broadband. Those who do use about 121 GB a month on it.
- They have about 0.4 late payments in six months.

Upgrade potential: rank 4 of 5 (score 19.3)

- Most already have a plan that fits. 28% have outgrown their plan (over their data or call allowance and not on the top plan), and only 10% go over on data.
- 20.5% upgraded in the last 24 months, which is above average.
- They rarely accept promotions (9.2%).

Points to keep in mind

- This segment contains a smaller group of about 730 customers (14% of the segment) who roam abroad on three or more days in three months. They spend about 42.5 a month against 30.7 for the rest, and they use

about 18 GB of data and 394 call minutes a month, against 10 GB and 255 minutes for the rest. A group like this formed its own segment in a six-segment test, but at 4.9% of customers it was just under the 5% minimum size rule.

Segment 2: Light Prepaid Users

27.0% of customers (4,051)

Summary. Prepaid customers with light use and low spending. Most are on the Basic plan, and they are the least responsive to promotions and the least likely to have upgraded.

Key fact	This segment	All customers
Average monthly spend	19.9	26.8
Mobile data per month	4.6 GB	11.3 GB
Top-ups per month	1.8	not applicable
Average value of one top-up	13.1	not applicable
Accepts promotions (average)	4.9%	12.5%
Upgraded in last 24 months	9.4%	17.6%

Who they are

- Every customer is prepaid.
- 53% are on the Basic plan and 31% on Standard.
- 31% are aged 18 to 34 and 17% are 55 and over.
- 52% live in urban areas.
- They mostly signed up in retail shops (41%) or through resellers (26%).

How they use TelcoX

- Low data (median 3.0 GB) and about 160 call minutes a month.
- About two top-ups a month, and each top-up is larger than in Segment 3.
- Few bundles (0.7 in three months).
- The typical customer uses about a third of their data allowance.

Upgrade potential: rank 5 of 5 (score 4.2)

- 34% have outgrown their plan (over their data or call allowance and not on the top plan), mostly on call minutes. Only 11% go over on data.
- Past upgrades and response to promotions are both the lowest of any segment.

Points to keep in mind

- Age is not stated for 18% of this segment, so the age shares above are lower than the true shares.

Segment 3: Deal-Seeking Prepaid Users

18.3% of customers (2,749)

Summary. Almost entirely prepaid, with frequent small top-ups and many bundle purchases. They respond to promotions more than any other segment, and they are the newest customers.

Key fact	This segment	All customers
Top-ups per month	4.7	not applicable
Bundles bought in 3 months	5.0	2.0
Accepts promotions (average)	27.4%	12.5%
Years as a customer	1.8	2.9
Aged 18 to 34	50%	37%

Key fact	This segment	All customers
Customers over their data allowance	42.5%	28.0%

Who they are

- 99.7% are prepaid.
- Half are aged 18 to 34, and only 5% are 55 and over.
- 58% live in urban areas.
- They sign up through a mix of channels: the app (29%), resellers (28%), and self-service (22%). Retail shops are used least (21%).
- Plans are spread out: 37% Basic, 38% Standard, and 25% Plus or Premium.

How they use TelcoX

- Many small top-ups, about 8.9 each.
- They send the most texts of any segment (137 a month) and buy the most bundles.
- Data use is average to high (12.3 GB, median 8.8 GB), and they went over their data limit in 1.2 of the last 3 months on average.
- Despite being prepaid, they spend almost as much as Segment 1 (31.9). The money comes in through many small top-ups and bundles.

Upgrade potential: rank 3 of 5 (score 54.6)

- 51% have outgrown their plan (over their data or call allowance and not on the top plan).
- 19.7% upgraded in the last 24 months.
- Their willingness to respond to offers is the strongest of any segment.

Points to keep in mind

- Age is not stated for 18% of this segment, so the youth share is probably higher than shown.
- They are the newest customers, so they have the shortest history with TelcoX.

Segment 4: Heavy Data Users

12.3% of customers (1,848)

Summary. The heaviest data users by a wide margin. Nearly all of them use more data than their plan allows, almost every month, and nearly all are on a Basic or Standard plan.

Key fact	This segment	All customers
Mobile data per month	28.8 GB	11.3 GB
Customers over their data allowance	98.6%	28.0%
On Basic or Standard plan	97%	73%
Average monthly spend	25.0	26.8
Upgraded in last 24 months	28.4%	17.6%
Aged 18 to 34	60%	37%

Who they are

- A mix of account types: 60% postpaid and 40% prepaid.
- The youngest segment: 60% are aged 18 to 34 and only 6% are 55 and over.
- The most urban segment (72%).
- The most likely to have signed up through the app (42%).
- 65% are on Basic, 31% on Standard, and almost none on Plus or Premium.

How they use TelcoX

- About 29 GB of data a month, which is more than two and a half times the average customer. The typical customer uses over three times their data allowance.
- They went over their data limit in nearly every month (2.9 of the last 3).
- Half of their usage happens at off-peak times, the highest share of any segment.
- Call use is below average (186 minutes) and text use is above average (112).

Upgrade potential: rank 1 of 5 (score 91.3)

- 98.6% use more than their plan allows and are not on the top plan.
- They have the highest rate of past upgrades (28.4%) and respond to promotions at a good level (20.0%).
- The top rank holds under equal weights. Under a ranking on need alone, they come second by a small margin.

Points to keep in mind

- Their spending (25.0) is below the all-customer average even though they use far more data. This suggests they sit on plans that are cheap for how much they use.

Segment 5: Long-Term Voice Users

8.1% of customers (1,215)

Summary. The smallest segment. These customers make about three times as many calls as the average customer, use little data, have been with TelcoX the longest, and skew older.

Key fact	This segment	All customers
Call minutes per month	757	252
Customers over their call allowance	100%	35.8%
Years as a customer	4.7	2.9
Aged 55 and over	33%	15%
Average monthly spend	17.7	26.8
Accepts promotions (average)	7.1%	12.5%

Who they are

- Mostly postpaid (73%).
- The oldest segment: 33% are aged 55 and over, and only 16% are aged 18 to 34.
- The least urban segment (50%).
- More than half signed up in retail shops (51%), the highest share of any segment.
- 59% are on Basic and 34% on Standard.

How they use TelcoX

- About 757 call minutes a month. The typical customer uses more than four times their call allowance.
- Low data use (median 3.7 GB), and the lowest share of off-peak use (26%).
- The lowest average spend of any segment (17.7).
- The fewest late payments (about 0.3 in six months).

Upgrade potential: rank 2 of 5 (score 56.1)

- They have the strongest need of any segment (99.3%), because nearly all exceed their call allowance.
- Past upgrades (12.2%) and response to promotions (7.1%) are low.
- Their rank depends on the weights. They come first on need alone, and third under equal weights.

Points to keep in mind

- A strong need combined with weak response suggests this group may be harder to reach through promotions than through other routes.

5. Using These Profiles With Care

- **The segments overlap.** Customer behavior changes gradually, so there are no clear gaps between groups. A customer near the edge of one segment could fit another. The silhouette score for five segments was 0.157, which is low. Use the segments as a practical way to divide customers, not as exact labels.
- **The segments are stable.** In 20 tests on random subsamples, at least 99.3% of customers stayed in the same segment.
- **Age is incomplete.** Age is not stated for 11% of all customers and for 18% of Segments 2 and 3, so age shares are understated there.
- **Region does not separate the segments.** Each region makes up almost the same share of every segment, so region is not used in the profiles.
- **Upgrade scores depend on the weights.** The weights are a judgment call. Segment 4 ranks first in two of the three weightings tested, and Segments 1 and 2 are always in the bottom two.
- **Names and numbers match the seed 42 data.** If the data is regenerated with a different seed, the numbering and numbers can change.
- **The data is synthetic.** These profiles show how the method works. They do not describe real TelcoX customers.

Marketing Recommendations

Project name	TelcoX Customer Segmentation
Document	08 Marketing Recommendations
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	03 Requirements and User Stories, 07 Segment Profiles, 09 KPI and Measurement Plan

1. Purpose of This Document

This document turns the five customer segments into practical marketing actions. For each segment it gives one main offer, one channel, and one timing, with the reason each one fits. It also includes a one-page guide for shop staff and a short briefing for customer care. It supports requirements FR12 to FR14 and user stories US9 to US11.

Scope. All offers use TelcoX's existing plans and channels. Designing new plans and setting new prices are outside this project. Where an offer needs a discount or bonus, the exact amount is for the product and pricing team to decide.

2. Principles Behind the Recommendations

1. **Target by need first.** Within every segment, upgrade offers go only to customers who have outgrown their plan. That means customers who use more data or call minutes than their plan allows and are not already on the top plan. There are 7,206 such customers, 48 percent of all customers.
2. **Rank describes concentration, not volume.** The lower-ranked segments still contain about 1,400 such customers each. A need-based filter finds them without sending offers to the rest of the segment.
3. **Match the channel to the segment.** Segments that rarely respond to promotions get personal contact. Segments that respond well get quick digital offers at the moment they top up or hit a limit.
4. **Fewer, better offers.** Customers received about five promotions in six months, and every segment received almost the same number. Offers should now differ by segment, and no customer should receive more than one upgrade or cross-sell offer in a month.
5. **Fit the plan to the customer.** Offer the smallest plan that covers the customer's actual use. A customer who is sent to a plan that is too large is more likely to refuse or to downgrade later.

3. Channels TelcoX Already Uses

Channel	How it is used
Retail shops	Face-to-face sales and service. Signing-up channel for many customers.
Mobile app	In-app messages and notifications, plan management, top-ups.
USSD and self-service	Balance checks, top-ups, and short menu offers on any phone.
SMS from the campaign tool	Offers and reminders sent to customers.
Resellers	Sell top-ups and sign up customers, mainly prepaid.
Customer care	Phone and support contact.

Every recommendation below uses only these channels (requirement FR13).

4. Summary of Recommendations

Segment	Customers to target	Main offer	Channel	Timing	Why it fits
4. Heavy Data Users	1,823 who have outgrown their plan	Move to the smallest plan that covers their data use, with a first-month incentive	Mobile app	The day they reach their data allowance	Nearly all exceed their allowance almost every month, and they are the most app-oriented segment.
5. Long-Term Voice Users	1,207 who have outgrown their plan	A free plan check-up that moves them to a plan covering their calls	Retail shop, with an SMS invitation	After a bill showing they went over their call allowance	They rarely respond to promotions, and half of them signed up in a shop.
3. Deal-Seeking Prepaid Users	1,392 who have outgrown their plan	Bonus data on a larger monthly bundle, as a step towards a monthly plan	USSD and self-service	At the moment they top up	They top up about 4.7 times a month and respond to offers more than any other segment.
1. Steady Postpaid Households	3,165 without home broadband	Home broadband and mobile bundle	Retail shop	At the next shop visit or customer care contact	They have the highest broadband take-up (38%), and they respond rarely to promotions, so a personal offer suits them.
2. Light Prepaid Users	1,364 who have outgrown their plan	A plan with a larger call allowance for those over their current call limit	USSD and self-service	At the moment they top up	Most of their need is on call minutes, and USSD is a low-cost way to reach a group that rarely responds.

5. Recommendations by Segment

5.1 Heavy Data Users (Segment 4)

Goal: move customers onto a plan that fits their data use.

Who to target. The 1,823 customers who use more data than their plan allows. Of the 1,848 customers in the segment, 731 are prepaid and 1,117 are postpaid. Prepaid customers should be offered the equivalent monthly plan or data bundle.

Main recommendation

- **Offer:** the smallest plan that covers their average data use, with a first-month incentive. The plan table makes the rule simple. Standard (15 GB) suits 14 percent of the segment, Plus (40 GB) suits 68 percent, and Premium (100 GB) suits the remaining 18 percent.
- **Channel:** mobile app, using an in-app notification.
- **Timing:** the day the customer reaches their data allowance.

Supporting actions

- Show the customer their own usage against their allowance, so the case for upgrading is clear.
- Send an SMS to customers who do not use the app. 42 percent of the segment signed up through the app, so most can be reached there.

Example message (in-app). "You used [31] GB of data last month, but your plan includes [5] GB. The Plus plan includes 40 GB. Tap to see the price and switch."

Why it fits. 98.6 percent are over their allowance, and they went over in 2.9 of the last 3 months. They have the highest rate of past upgrades (28.4%).

What to avoid. Call or text offers. This segment's call use is below average. Also avoid sending the offer in months when the customer was within their limit.

5.2 Long-Term Voice Users (Segment 5)

Goal: move heavy callers onto a plan that covers their calls, and thank them for their long relationship.

Who to target. The 1,207 customers who use more call minutes than their plan allows. 884 of the 1,215 in this segment are postpaid.

Main recommendation

- **Offer:** a free plan check-up. The agent finds the plan whose call allowance covers the customer's use. By the plan table, Standard (300 minutes) suits 1 percent, Plus (600 minutes) suits 34 percent, and Premium (1,000 minutes) suits 45 percent. The remaining 20 percent use more than 1,000 minutes a month, which is above even the Premium allowance, so they need an extra call-minutes bundle on top of their plan.
- **Channel:** retail shop, with an SMS invitation to visit.
- **Timing:** after a bill showing that the customer went over their call allowance. For prepaid customers, after a top-up that follows a month of heavy calling.

Supporting actions

- Customer care can offer the same check-up when these customers call in.
- Use a thank-you tone that mentions how long they have been customers.

Example message (SMS). "Thank you for [5] years with TelcoX. You made about [750] call minutes last month. Visit any TelcoX shop for a free plan check-up and we will find the plan that fits your calls."

Why it fits. 99.3 percent are over their call allowance, yet only 7.1 percent respond to promotions. A personal conversation in a shop suits that pattern. They are also the oldest segment, with 33 percent aged 55 and over.

What to avoid. Data offers, because they use little data (median 3.7 GB). Also avoid pushing the largest plan. Their average spend is the lowest of any segment (17.7), so a big price jump may be refused. Starting from the smallest plan that fits is more likely to succeed.

5.3 Deal-Seeking Prepaid Users (Segment 3)

Goal: turn frequent small purchases into a steadier, higher-value plan.

Who to target. The 1,392 customers who have outgrown their plan. 1,168 of them go over their data allowance.

Main recommendation

- **Offer:** bonus data when the customer buys a larger monthly bundle, as a step towards a monthly plan. The offer should be time-limited, because this segment is used to responding to deals.
- **Channel:** USSD and self-service.
- **Timing:** at the moment they top up, which happens about 4.7 times a month.

Supporting actions

- Use SMS and in-app messages on the same day as the top-up offer.
- Resellers sell a lot of top-ups to this group (28% signed up through resellers). Resellers can mention the offer at the point of sale.

Example message (USSD). "Top up [10] or more and get bonus data on a monthly bundle. Reply 1 to see the offer."

Why it fits. They respond to promotions at 27.4 percent, which is the highest of any segment. They buy about 5 bundles in three months and go over their data limit in 1.2 of the last 3 months.

What to avoid. Offers every week, because this group may learn to wait for the next deal. Also avoid long contracts. They are the newest customers (1.8 years on average) and prefer small commitments.

5.4 Steady Postpaid Households (Segment 1)

Goal: grow the value of a loyal, stable segment by adding services, instead of pushing upgrades.

Who to target. The 3,165 customers without home broadband. Two smaller groups can be handled as extra actions: 733 customers who roam abroad on three or more days in three months, and 1,420 customers who have outgrown their plan.

Main recommendation

- **Offer:** a home broadband and mobile bundle.
- **Channel:** retail shop.

- **Timing:** at the next shop visit or customer care contact.

Supporting actions

- Roaming customers: offer a roaming add-on when their first roaming day is detected. They spend about 42.5 a month against 30.7 for the rest of the segment.
- Customers who have outgrown their plan: offer a plan check-up at the same visit.

Example script (shop). "Many of our long-standing customers have added home broadband to their mobile plan. Would you like to hear what it would cost for you?"

Why it fits. This segment has the highest broadband take-up (38.4% against 21.9% overall), the longest-standing postpaid relationships (3.5 years), and a low response to promotions (9.2%). A personal offer at a moment when the customer is already in contact is more likely to work than a broad message.

What to avoid. Broad SMS blasts. This is the largest segment, and a low response rate would waste a lot of spend. Most of its customers (72%) are within their plan limits, so an upgrade push does not fit.

5.5 Light Prepaid Users (Segment 2)

Goal: serve the small group with a clear need at low cost, and keep the rest lightly engaged.

Who to target. The 1,364 customers who have outgrown their plan. Most of them are over their call allowance (1,152 customers), and 799 of those are on the Basic plan with its 100-minute limit.

Main recommendation

- **Offer:** a plan with a larger call allowance, such as Standard (300 minutes), for customers over their current call limit.
- **Channel:** USSD and self-service.
- **Timing:** at the moment they top up.

Supporting actions

- Keep contact light for the rest of the segment. Do not run broad upgrade campaigns here.
- Resellers (26% of the segment) can pass on the offer in person.

Example message (USSD). "You used [250] call minutes last month. The Standard plan includes 300. Reply 1 to learn more."

Why it fits. The typical customer uses only about a third of their data allowance, so the need is on calls. Response to promotions is the lowest of any segment (4.9%), so a low-cost channel keeps the spend small.

What to avoid. Data offers and expensive campaigns. This segment spends little (19.9) and rarely upgrades (9.4%), so heavy investment is unlikely to pay back.

6. Campaign Priority and Order

Order	Segment	Customers targeted	Reason
1	Heavy Data Users	1,823	Clearest need and the highest upgrade potential. Quick to run in the app.
2	Deal-Seeking Prepaid Users	1,392	Best response rate, and the offer reaches them at the moment of top-up.
2	Long-Term Voice Users	1,207	Strongest need, but a slower, personal approach through shops. Run at the same time as the deal-seekers.
3	Steady Postpaid Households	3,165	Growth through broadband. Largest group, so start with shop-based offers and review results before widening.
3	Light Prepaid Users	1,364	Lowest priority. Run a small, low-cost campaign.

The order follows the upgrade ranking in document 07, with segment 1 treated as a cross-sell campaign. Document 09 explains how the campaigns will be tested before they are widened.

7. Retail Quick Guides

One short guide per segment for shop staff (requirement FR14, user story US10). Staff can see a customer's plan and usage when they look up the account.

Heavy Data Users

- **How to recognize them:** uses far more data than the plan allows, often on a Basic or Standard plan.
- **Opening line:** "I can see you have been using a lot of data. Let me show you a plan that would cover it."
- **Offer:** the smallest plan that covers their data, with the first-month incentive.
- **Avoid:** talking about calls or texts.

Long-Term Voice Users

- **How to recognize them:** a long-standing customer with call minutes well above the plan allowance.
- **Opening line:** "Thank you for being with us so long. Let's check whether your plan still suits the way you use it."
- **Offer:** the plan check-up, moving to the plan that covers their calls. Add an extra call-minutes bundle for those over 1,000 minutes.
- **Avoid:** pushing the biggest plan, or leading with price.

Deal-Seeking Prepaid Users

- **How to recognize them:** top-ups several times a month, many bundles, usually younger.
- **Opening line:** "I can see you often buy bundles. There is a bigger monthly bundle with bonus data that may save you trips."
- **Offer:** bonus data on a larger monthly bundle.
- **Avoid:** long contracts.

Steady Postpaid Households

- **How to recognize them:** postpaid for several years, on a mid-range plan, within their limits.
- **Opening line:** "Many of our long-standing customers have added home broadband. Would you like to hear what it would cost for you?"
- **Offer:** home broadband and mobile bundle. Mention the roaming add-on to customers who travel.
- **Avoid:** upgrade pressure. Their plan usually fits.

Light Prepaid Users

- **How to recognize them:** low use, small top-ups, Basic plan.
- **Opening line:** "I can see you make quite a few calls. A plan with more call minutes might suit you."
- **Offer:** a plan with a larger call allowance, only if they exceed their current limit.
- **Avoid:** data offers.

8. Customer Care Briefing

Likely questions after each campaign (requirement US11). In every case, the first step is to show the customer their last three months of usage next to their plan allowance.

Segment	Likely questions	How to help
Heavy Data Users	"Why did I run out of data so quickly?" "Do I lose my remaining data if I switch part-way through the month?"	Explain allowance against usage. Check the switching rules before promising anything.
Long-Term Voice Users	"Can I keep my number and my current plan if I say no?" "Is there a plan that covers all of my calls?"	Confirm that declining changes nothing. Explain the call allowance of each plan, and the extra call-minutes bundle.

Segment	Likely questions	How to help
Deal-Seeking Prepaid Users	"What happens to my bonus data when it expires?" "Do I have to keep topping up to keep the offer?"	State the end date and conditions of the bonus clearly.
Steady Postpaid Households	"Is broadband available at my address?" "Will adding broadband change my mobile bill?"	Check availability first. Explain what is billed together and what is separate.
Light Prepaid Users	"Why am I being offered a plan?" "Will my top-ups work the same way?"	Show their call use against the Basic allowance, and explain that top-ups work as before.

9. Requirements Covered

Requirement	Where
FR12 Offer, channel, and timing for each segment	Sections 4 and 5
FR13 Every channel is one TelcoX already uses	Section 3
FR14 One-page guide for sales and retail staff	Section 7
US9 Tailored offers, with a one-sentence reason	Sections 4 and 5
US10 Retail guide with description, offer, and conversation starter	Section 7
US11 Likely customer care questions, at least two per segment	Section 8

KPI and Measurement Plan

Project name	TelcoX Customer Segmentation
Document	09 KPI and Measurement Plan
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	01 Business Requirements, 03 Requirements and User Stories, 08 Marketing Recommendations

1. Purpose of This Document

This document explains how TelcoX will know whether segment-based marketing works. It defines the measures of success, shows where each one starts today, describes a fair test against the current approach, and sets out the rule for deciding what happens next. It also covers how often the segments should be refreshed. It supports requirements FR17, FR18, and FR20 and user stories US13, US14, and US16.

The BRD left target numbers open and promised they would be set here. Section 6 proposes them.

2. Goals and the Measures That Track Them

Business goal (document 01)	Measure that tracks it
B1 More customers upgrade to higher plans	Upgrade rate
B2 More customers respond to promotions	Response rate
B3 Less money spent on offers that do not fit	Cost per extra upgrade
B4 Higher ARPU, especially where needs are unmet	ARPU change
B5 One shared view of customer groups	Segment use by teams

3. The Measures

3.1 Main measures

ID	Measure	Plain meaning	How it is calculated	Source	How often
K1	Upgrade rate	The share of customers who moved to a higher plan	Customers who upgraded divided by customers in the group	Billing and customer database	Weekly during the campaign, final result at 3 months
K2	Response rate	The share of offers that customers accepted	Offers accepted divided by offers sent	Campaign tool	Weekly
K3	ARPU change	How much average monthly spend per customer changed	Average monthly spend at the end of the test minus average spend in the month before the campaign. Compared between the targeted and control groups.	Billing	Monthly
K4	Cost per extra upgrade	What each additional upgrade cost	Total campaign cost divided by the extra upgrades in the targeted group (its upgrades minus the upgrades the control group would predict)	Finance and campaign tool	At the end of the test

3.2 Supporting measures

ID	Measure	Plain meaning	How it is calculated
K5	Broadband take-up (Segment 1)	The share of customers offered broadband who added it	Customers who added broadband divided by customers offered
K6	Plan fit	Whether targeted customers still use more than their plan allows	Share of targeted customers over their allowance, before and after. It should fall.
K7	Segment use	Whether teams actually use the segments	Share of campaigns planned using segments, checked each quarter
K8	Segment stability	Whether the segments hold up over time	Share of customers who stay in the same segment at each refresh

3.3 Guardrail measures

These make sure the campaigns do not cause harm. They are compared between the targeted and control groups.

ID	Measure	How it is calculated
G1	Opt-out rate	Customers who opted out of messages divided by customers contacted
G2	Complaint rate	Complaints per 1,000 customers contacted
G3	Disconnection rate	Customers who left TelcoX divided by customers in the group

4. Where We Start: Baselines

The baseline is how the current broad approach performs. These figures come from the customer data. In a real project, they would come from the campaign tool and billing records for the last 6 to 24 months.

Segment	Response rate (offers accepted ÷ sent, last 6 months)	Monthly upgrade rate (upgrades in last 24 months ÷ 24)	Average monthly spend
1. Steady Postpaid Households	9.3%	1.00%	32.4
2. Light Prepaid Users	5.2%	0.41%	19.9
3. Deal-Seeking Prepaid Users	26.8%	0.93%	31.9
4. Heavy Data Users	20.3%	1.40%	25.0
5. Long-Term Voice Users	6.8%	0.58%	17.7
All customers	12.6%	0.84%	26.8

- Among customers who have outgrown their plan, the monthly upgrade rate is about 0.88 percent.
- **Cost per extra upgrade has no baseline in the data.** The finance team would supply the current cost of a campaign and its upgrades.
- **Why the response rate differs from document 07.** Document 07 averages each customer's own rate. This table adds up all offers accepted and divides by all offers sent. The two are close (12.5% and 12.6%).

5. A Fair Test

A fair test compares customers who got the new approach with similar customers who got the usual one. Without it, any change in upgrades could be put down to the season, a price change, or a competitor.

5.1 Steps

1. **List the eligible customers** in each segment. For the upgrade campaigns (Segments 2 to 5), these are customers who use more data or call minutes than their plan allows and are not on the top plan. For the Segment 1 broadband campaign, these are customers without home broadband.

2. **Split each list at random into two equal halves.** The split is done inside each segment, so the two halves are alike. A fixed random seed makes the split repeatable.
3. **Targeted group:** receives the segment offer from document 08.
4. **Control group:** receives the usual broad offer, as today.
5. **Lock the lists.** No customer moves between groups once the campaign starts.
6. **Run the campaign for 4 weeks.**
7. **Keep counting upgrades for 8 more weeks.** The total measuring period is 3 months, because customers do not always upgrade the day they receive an offer.
8. **Compare the groups** on every measure in section 3 and apply the decision rule in section 7.

Why half and half. The control group still gets an offer, so no customer is left out. An equal split also gives the most sensitive test.

5.2 How big a change can the test detect?

Upgrades are rare. The average customer has less than a 1 percent chance of upgrading in a given month, so small groups cannot show small improvements. The table shows the smallest increase over the control group that the test could reliably tell apart from chance, using a standard 5 percent significance level and 80 percent power.

Group	Eligible customers	Customers per half	Control group's expected upgrade rate over 3 months	Smallest lift the test can detect
4. Heavy Data Users	1,823	911	4.1%	63%
3. Deal-Seeking Prepaid Users	1,392	696	2.8%	89%
5. Long-Term Voice Users	1,207	603	1.7%	123%
2. Light Prepaid Users	1,364	682	1.3%	130%
1. Steady Postpaid Households (plan check-up)	1,420	710	2.6%	92%
Segments 2 to 5 pooled	5,786	2,893	2.6%	45%

How to read it. For Heavy Data Users, the control group would be expected to upgrade at about 4.1 percent. A lift of 63 percent means the targeted group would need to reach about 6.7 percent before the test could call it a real difference.

What this means for the design

- **The formal decision uses the pooled result for Segments 2 to 5.** Pooling the four upgrade campaigns gives groups large enough to detect a 45 percent lift over three months.
- **Segment results are shown but treated as directional.** Single segments are too small to detect modest lifts, except possibly Heavy Data Users.
- **A longer test helps.** For the pooled group, the smallest detectable lift falls from 78 percent at one month to 45 percent at three months and 31 percent at six months. If the three-month result is unclear, the test should continue to six months.
- **Response rate is easier to measure than upgrade rate.** Offers are accepted far more often than plans change, so the response rate will give an early reading while the upgrade rate catches up.

6. Proposed Targets

These are proposals for leadership to confirm. Each is set so that the test can actually detect it.

Measure	Proposed target
Upgrade rate	At least 45% higher than the control group over three months, in the pooled test. For example, from 2.6% to about 3.8%.
Response rate	At least 25% higher than the control group's response rate
ARPU change	Higher in the targeted group than in the control group

Measure	Proposed target
Cost per extra upgrade	At or below the break-even level in section 7.2
Broadband take-up (Segment 1)	Significantly higher than the control group. The first test sets the baseline.
Guardrails (G1 to G3)	No significant increase compared with the control group

7. Decision Rule

7.1 When to widen the campaigns

Widen the segment-based campaigns to all customers in the segment if **all** of these are true at the end of the test:

1. The upgrade rate in the targeted group meets the target in section 6, and the difference is statistically significant (5 percent level).
2. The response rate meets its target.
3. ARPU change is higher in the targeted group than in the control group.
4. Cost per extra upgrade is at or below break-even.
5. None of the guardrail measures is significantly worse in the targeted group.

If the upgrade rate is promising but not yet significant, extend the test to six months. If a guardrail is significantly worse, stop and review the offer before continuing.

7.2 Break-even cost

An upgrade is worth running if the extra money it brings in over the months the customer stays is more than the cost of getting it.

Break-even cost per extra upgrade = extra monthly spend after the upgrade × months the customer is expected to stay

Example. Moving from Standard (20) to Plus (35) adds 15 per month. If the customer stays for 12 months, the extra revenue is 180. A cost per extra upgrade below 180 would pay back. The finance team should replace revenue with margin if it is available.

8. Reporting

Report	Audience	How often	Content
Campaign dashboard	Marketing	Weekly during the campaign	Response rate, upgrades so far, opt-outs and complaints, targeted against control
Test result summary	Leadership, product, marketing	At 3 months (and 6 months if extended)	All measures, target comparison, decision
Segment health check	Marketing, data team	Every quarter	Segment sizes, stability, and use of segments in campaigns

9. Keeping the Segments Up to Date

Recommended refresh: every quarter. Customer behavior changes gradually, and campaigns change it further. For example, Heavy Data Users who upgrade will use less of their allowance and may fit a different segment next time.

How to refresh. Rerun the segmentation notebook on the latest data, then compare the new segments with the previous ones in size and profile and check whether the names still fit.

Signs the segments are out of date

- Any segment's share of customers has moved by more than 5 percentage points since the last refresh.
- The stability check falls below 95 percent. The current result is at least 99.3 percent.

10. Requirements Covered

Requirement	Where
FR17 Define the measures of success and how each is calculated	Section 3
FR18 A test comparing targeted customers with a control group	Section 5
FR20 Recommend how often segments should be refreshed	Section 9
US13 Each measure is defined, calculated, and compared with the current approach	Sections 3, 4, and 6
US14 A fair test with groups, selection steps, duration, and decision rule	Sections 5 and 7
US16 Refresh schedule with one sign of being out of date	Section 9

Process Flow

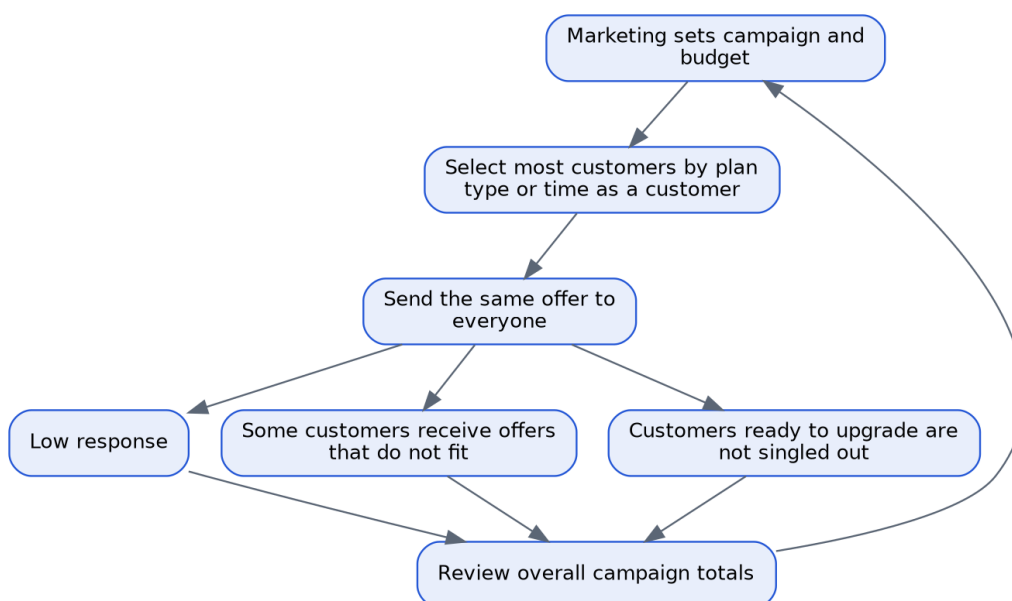
Project name	TelcoX Customer Segmentation
Document	11 Process Flow (optional deliverable)
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	01 Business Requirements, 08 Marketing Recommendations, 09 KPI and Measurement Plan

1. Purpose of This Document

This document shows how TelcoX runs marketing campaigns today and how the process changes with customer segments. The diagrams use Mermaid, which GitHub displays as pictures when the file is opened in the repository.

2. Current Process: Broad Campaigns

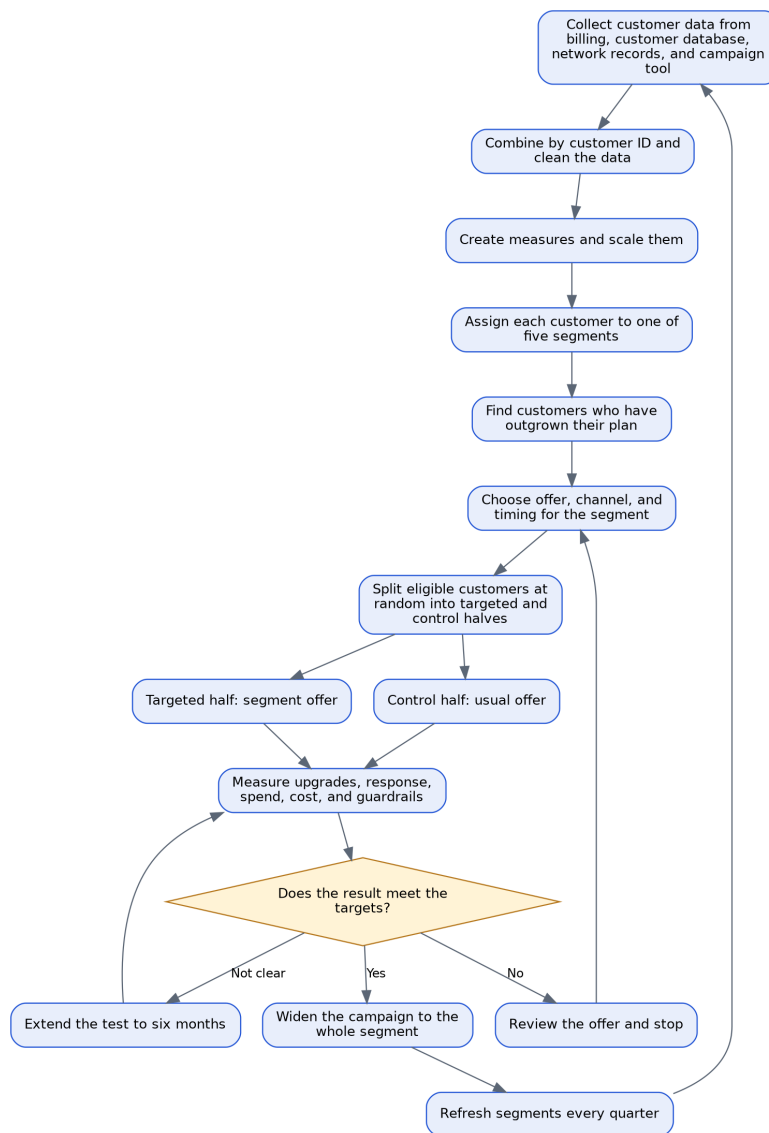
Today, most customers receive the same offers, and results are judged by overall totals.



What goes wrong: response is low, spending is high, and nobody learns which customers were ready to upgrade.

3. Proposed Process: Segment-Driven Campaigns

The proposed process uses customer behavior to decide who gets which offer, and tests the result against the usual approach.



4. Who Does What

Step	What happens	Owner	Supporting team
Collect and combine data	Pull the four data sources into one customer table	IT and data team	Business systems analyst
Clean and prepare	Fix problems, create measures, scale	Business systems analyst	IT and data team
Assign segments	Run the segmentation and save each customer's segment	Business systems analyst	Data team
Find customers who have outgrown their plan	Apply the need rule	Business systems analyst	Product and pricing team
Choose offer, channel, and timing	Use document 08 and agree the incentive amounts	Marketing	Product and pricing team
Check privacy	Review the data fields and who can see segment labels	Legal and compliance team	Data team
Split and send	Create the two groups and run the campaign	Marketing	IT and data team
Brief front-line staff	Share shop guides and care briefing	Sales and retail, customer care	Marketing
Measure and decide	Report results and apply the decision rule	Business systems analyst	Marketing, leadership

Step	What happens	Owner	Supporting team
Refresh	Rerun the segmentation each quarter	Business systems analyst	Data team

5. What Changes

	Today	With segments
Who gets an offer	Most customers	Customers who match their segment and have outgrown their plan
What they receive	The same offer	An offer built for their segment
Channel and timing	Whatever suits the campaign	The channel and moment that fit the segment
How results are judged	Overall totals	Targeted group compared with a control group
How often it is reviewed	After each campaign	Each campaign, with segments refreshed every quarter

Risk and Limitations Log

Project name	TelcoX Customer Segmentation
Document	12 Risk and Limitations Log (optional deliverable)
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	01 Business Requirements, 05 Data Quality and Preparation Report, 09 KPI and Measurement Plan

1. Purpose of This Document

This log lists what could go wrong with the project and its results, how likely and how serious each risk is, and what reduces it. It also lists the limits of the work, so that no one reads more into the results than they can support. It builds on the risk table in document 01.

How to read the ratings. Likelihood and impact are rated High, Medium, or Low. These are judgments made for this case study, and the project team would review them with stakeholders.

2. Risk Log

ID	Risk	Type	Likelihood	Impact	How it is reduced	Owner
R1	The segments overlap, so customers near the edges may be treated as if they clearly belong to one group. The silhouette score for five segments is low (0.157).	Analysis	High	Medium	Present segments as a practical way to divide customers, not exact labels. Use the need rule so offers go to customers with a clear gap between use and plan.	Business systems analyst
R2	Customer behavior changes and the segments go out of date. Campaigns also change behavior, as when Heavy Data Users upgrade.	Analysis	High	Medium	Refresh every quarter. Watch segment size shifts of more than 5 points and a stability score below 95 percent.	Business systems analyst
R3	The upgrade ranking depends on the chosen weights (50, 25, 25). A different weighting moves Voice Users and Deal-Seekers between ranks 2 and 3.	Analysis	Medium	Medium	Show the ranking under other weights. Agree the weights with marketing and product before the pilot.	Business systems analyst
R4	The test is too small to detect modest improvements. Single segments could only detect lifts of 63 to 130 percent over three months.	Measurement	High	High	Pool Segments 2 to 5 for the formal decision (45 percent detectable lift). Extend to six months if unclear.	Business systems analyst
R5	Offers move customers to plans that are too large, leading to complaints or later downgrades.	Business	Medium	High	Offer the smallest plan that covers the customer's use. Track complaints and disconnections as guardrails.	Marketing
R6	Customers learn to wait for deals, especially Deal-Seeking Prepaid Users.	Business	Medium	Medium	Time-limited offers, no weekly offers, and a limit of one upgrade or cross-sell offer per customer per month.	Marketing
R7	Too many messages cause opt-outs and complaints.	Business	Medium	Medium	Contact limit per customer. Report opt-out and complaint rates in every campaign review.	Marketing
R8	Teams do not adopt the segments, or keep defining high-value customers in their own ways.	Adoption	Medium	High	Short plain-language profiles, shop guides, and a care briefing. Agree one shared definition of a high-value customer at the start.	Marketing lead
R9	Real customer data is messier than the synthetic data, with more gaps, mismatched customer IDs between systems, and inconsistent units.	Data	High	Medium	Run the full quality checks in document 05 on real data first. Add checks for dates, units, and ID matching.	IT and data team

ID	Risk	Type	Likelihood	Impact	How it is reduced	Owner
R10	Sensitive personal details are used or exposed. Gender is the only sensitive field in the case-study data.	Privacy	Low	High	Keep personal details out of the analysis and keep gender out of the grouping. Review fields and access with the legal and compliance team.	Legal and compliance
R11	Segment labels are seen by people who should not see them, or are used for purposes other than marketing.	Privacy	Low	Medium	Limit access to the label table and record who uses it and why.	Data team
R12	A recommended channel or service is not available. For example, broadband may not reach some Segment 1 customers.	Business	Medium	Medium	Check availability before making the offer, and screen eligible customers by coverage.	Product and pricing
R13	Cleaning choices, such as limiting values at the 99th percentile and treating empty fields as 0, affect the segments.	Analysis	Medium	Low	The choices are documented. A future check could show how much the segments change under other choices.	Business systems analyst
R14	Results from a three-month test are read as permanent.	Measurement	Medium	Medium	Report the test period with each result and repeat measurement after refreshes.	Business systems analyst

3. Limitations

These are not risks to manage. They are limits of what the project has done.

- **The data is synthetic.** The case study shows how the method works and does not describe real customers. The groups in the data were designed, so the match between segments and real customer types has not been tested.
- **No cause and effect has been shown yet.** The segments describe behavior. Whether the offers raise upgrades is only known after the fair test in document 09.
- **Churn is not modelled.** Customers leaving TelcoX is outside the scope. It is only watched as a guardrail.
- **Prices and plans are unchanged.** The project works within existing plans. Whether the plans themselves are right is a separate question for the product and pricing team.
- **Cost data is missing.** Cost per extra upgrade has no baseline until finance supplies campaign costs.
- **Some segment details are incomplete.** Age is not stated for 11 percent of customers, and for 18 percent in Segments 2 and 3.
- **The roaming group in Segment 1 is not a segment of its own.** A six-segment test showed that a group like it would be just under the 5 percent minimum size.

4. Review Schedule

When	What
Before the pilot	Review R3, R4, R5, R7, R8, and R12 with stakeholders
At each campaign review	Check R5, R6, and R7 with the guardrail measures
At each quarterly refresh	Review R1, R2, and R13, and update ratings
When moving to real data	Review R9, R10, and R11 first

Implementation and Maintenance Plan

Project name	TelcoX Customer Segmentation
Document	13 Implementation and Maintenance Plan (optional deliverable)
Prepared by	Walusungu Nyirenda
Role	Business Systems Analyst (case study)
Related documents	08 Marketing Recommendations, 09 KPI and Measurement Plan, 11 Process Flow, 12 Risk and Limitations Log

1. Purpose of This Document

This document explains how TelcoX would move from the analysis to running segment-based campaigns, and how the segments would be kept accurate afterwards. It supports requirement BR5 (segments are easy to reach and use) and BR9 (a review schedule). The timings are an indicative plan for a project of this size.

2. Plan at a Glance

Phase	What happens	Indicative length	Result
0. Approval	Confirm targets, privacy approach, and the open questions in section 6	2 weeks	Agreed scope and targets
1. Data and segments	Combine the real data, run the quality checks, build and review the segments	4 weeks	Segments agreed with stakeholders
2. Preparation	Load segment labels into the campaign tool, brief shops and customer care, set up reporting	2 weeks	Campaign ready to launch
3. Pilot	Run the fair test: 4 weeks of campaign, then 8 weeks of follow-up	12 weeks	Test results
4. Decision	Apply the decision rule from document 09	1 week	Widen, extend, or stop
5. Rollout	Widen campaigns in the order set in document 08	About 8 weeks	Segment campaigns running
6. Maintain	Refresh segments every quarter and review results	Ongoing	Up-to-date segments

3. Phase Details

Phase 0. Approval

- Leadership confirms the proposed targets in document 09.
- Legal and compliance review the data fields and who may see segment labels.
- Marketing and product agree the weights for the upgrade ranking and the incentive amounts.
- The team agrees one shared definition of a high-value customer.

Phase 1. Data and segments

- The IT and data team join billing, customer database, network, and campaign data using the customer ID.
- Run the checks from document 05 on the real data and record what is found.
- Rerun the segmentation notebook. Check whether the number of segments and the segment profiles still make sense. The five segments and names in this case study belong to the synthetic data and may change.
- Review the profiles with marketing, sales, and product before anything is used.

Phase 2. Preparation

- Create a segment table with one row per customer.

- Load the table into the campaign tool.
- Brief shop staff and customer care using the guides in document 08.
- Build the weekly campaign report from document 09.

Phase 3. Pilot

- Follow the fair test steps in document 09: eligible list, random split inside each segment, targeted and control groups, locked lists.
- Run the campaign for four weeks, then keep counting upgrades for eight more weeks.
- Report weekly on response, opt-outs, and complaints.

Phase 4. Decision

- Apply the decision rule. Widen if all conditions are met. Extend to six months if the upgrade result is promising but unclear. Stop and review the offer if a guardrail is worse.

Phase 5. Rollout

- Widen in this order: Heavy Data Users, then Deal-Seeking Prepaid Users and Long-Term Voice Users, then Steady Postpaid Households and Light Prepaid Users.
- Keep the contact limit of one upgrade or cross-sell offer per customer per month.

4. The Segment Table

The campaign tool needs a simple table. Each row describes one customer.

Field	Meaning
customer_id	The customer's ID
segment	Segment number
segment_name	Plain name of the segment
outgrown_plan	Yes if the customer uses more data or calls than the plan allows and is not on the top plan
recommended_plan	The smallest plan that covers the customer's use
offer_code	The offer for this segment (from document 08)
test_group	Targeted or control, during the pilot

The table holds no names, phone numbers, or other personal details. It is refreshed with each quarterly update.

5. Roles

Role	Responsibility
Business systems analyst	Runs the segmentation, builds the segment table, reports results
IT and data team	Provides and combines the data, loads the table into the campaign tool
Marketing	Plans and runs campaigns, owns the offers
Product and pricing	Agrees incentives, checks plan and service availability
Sales and retail, customer care	Use the guides and report customer feedback
Legal and compliance	Approves data use and access
Leadership	Approves targets and the final decision

6. Questions to Settle Before Starting

Question	Why it matters	Who answers
Can TelcoX load a segment label for each customer into the campaign tool?	The pilot cannot run without it.	IT and data team
Is home broadband available to the customers in Segment 1 who do not have it?	The Segment 1 offer depends on it.	Product and pricing
What are the incentive amounts for each offer?	Needed for the campaigns and for break-even costs.	Product and pricing, finance
What does a campaign cost today?	Needed for the cost per extra upgrade baseline.	Finance
Which customer details may be used?	Needed before the real data is analyzed.	Legal and compliance

7. Maintenance

Task	How often	Owner
Refresh the segments and compare with the last version	Every quarter	Business systems analyst
Check segment sizes and stability against the warning signs in document 09	Every quarter	Business systems analyst
Check whether segment names and profiles still fit	Every quarter	Business systems analyst, marketing
Review the risk log	Every quarter	Project team
Check that campaigns use the segments	Every quarter	Marketing lead
Review the data fields and access rights	Every year, or when fields change	Legal and compliance

Handing over. When the work moves from the analyst to the data team, hand over the notebook, the data dictionary, the segment table definition, and the quarterly checklist above. The notebook is built to run again from start to finish and gives the same results with the same data (requirement NFR2).

8. Checkpoints

When	Check
End of Phase 0	Targets, weights, and privacy approach approved
End of Phase 1	Stakeholders can explain each segment in one sentence (requirement NFR1)
End of Phase 2	Segment table loaded, and front-line staff briefed
End of Phase 3	Test finished with locked lists and complete data
End of Phase 4	Decision recorded with the reasons
Each quarter	Refresh completed and warning signs checked